BLOCK 2

DATA COLLECTION AND PROCESSING

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Block 2 on Data Collection and Processing consists of three units.

Unit 4, "Research Design Formulation," provides an overview of research designs and how they are used in marketing research. The major design categories of exploratory, descriptive, quasi-experimental, and experimental research are explained. The key considerations for adopting a specific research design and subtypes within the major categories are also discussed.

Unit 5, "Data Collection: Qualitative and Quantitative," discusses the main data types used for marketing research in the context of the various research designs. It also distinguishes between primary and secondary sources of data. The two fundamental methods for collecting data, namely communication and observation, are discussed along with the different variations. The unit also explains the sources of error in data collection.

Unit 6, "Data Processing," discusses various aspects of data processing, including editing, coding, classification, and the presenting of data in tables and graphs.

UNIT 4 RESEARCH DESIGN FORMULATION

Learning Outcomes

After studying this unit, you should be able to:

- explain the concept and importance of research design in the context of marketing research problems
- discuss the causality factor in the context of research design
- describe the various types of research designs
- apply the learned concept in different research situations.

Structure

4.1 INTRODUCTION

A research design is a framework or blueprint for conducting a marketing research project.

It details the procedures necessary for obtaining the information needed to structure or solve marketing research problems. Although a broad approach to the problem has already been developed, the research design specifies the details—the nuts and bolts—of implementing that approach. A research design lays the foundation for conducting the project. A good research design will ensure that the marketing research project is conducted effectively and efficiently. In this unit, various types of research design are discussed along with the marketing situations in which they could be most effectively used.

4.2 RESEARCH DESIGN-MEANING AND IMPORTANCE

The research design is a detailed blueprint for the research study that will be conducted and provides a general outline of the procedures that will be used. The purpose of a research design is to make sure that the necessary data for the problem at hand is precisely and efficiently collected. Simply said, it is the framework for the research study, a map that directs data collection and analysis. Typically, a research design involves the following components or tasks:

1. Define the information needed
2. Design the exploratory, descriptive, and/or causal phases of the research
3. Specify the measurement and scaling procedures
4. Construct and pre-test a questionnaire (interviewing form) or an appropriate form for data collection
5. Specify the sampling process and sample size
6. Develop a plan for data analysis

Depending on the purpose of the researcher, the research design may be a very detailed statement or it may merely provide the bare minimum of data needed to plan the research study. A research design should provide at least the following information to be effective:

a) A statement of the objectives of the study or the research output.

b) A statement of the data inputs required on the basis of which the research problem is to be solved.

c) The analytical methods to be applied to the data inputs in order to process and analyse them.

Let us use an illustration to understand the elements:

A recently opened supermarket offers a wide selection of products, from food to kitchenware and household appliances. The general manager of sales thinks that increasing people's purchases in larger quantities can increase the supermarket's overall profitability, which might be done by providing all customers with attractive cash discounts. The other executives have their doubts about this. Conducting a marketing research activity is one way to test the hypothesis.

1) The objective is to determine the margin earned on sales when this discount is offered and compare it with the margin earned when the discount is not offered.

2) The data to be collected over a period of time

   a) Sales in rupees to a selected sample of customers during the period when the discount is offered.

   b) Sales in rupees to the same customer when the discount is not being offered.

   c) The average order size in the two periods.
   d) The average margins earned during the two periods.
   e) The cost of promotional inputs regarding the discount.
3) The analysis of the data will be
   a) Sales in rupees in period I minus those in period II.
   b) Subtract cost of incentives.
   c) Also subtract cost of promotional inputs.

The significance of research design is reflected in the fact that it specifies what must be done and how it must be done in order to accomplish the research objectives. It expresses what is anticipated from the research exercise in terms of outcomes and the analytical input required to transform data into research findings. The activities that would need to be carried out in order to accomplish the research objective are clearly identified by the research design. Therefore, it is beneficial if the research design is documented once it has been finalised in order to give the researcher a frame of reference and avoid the study from straying.

The research design aids in providing direction to the computation and interpretation processes at the analysis and interpretation stages as well, enabling the development of recommendations and solutions. This is not meant to imply that a design is a fixed constraint that a study must always abide by, though. The research design simply reflects an expectation of likely outcomes; but, as the investigation progresses, many unexpected outcomes may surface, necessitating the formulation of new hypotheses or at the very least, changing some existing ones. The research design should not be seen as a restricting framework for a study but rather as a guidance.

4.3 CAUSALITY: THE BASIS OF CLASSIFICATION OF VARIOUS TYPES OF RESEARCH DESIGNS

There are four distinct research design categories, which can be broadly classified as follows:

1) Exploratory Designs
2) Descriptive Designs
3) Quasi-Experimental Designs
4) Experimental Designs

These designs will be discussed in detail very shortly. However, it should be noted that causality serves as the foundation for categorising various types of research designs. It is what is being discussed in this section.

Let us start our discussion with an example. Consider a scenario where the sales manager of an air conditioner manufacturer conducts a training programme for its salesmen working in a state. It was noted that there was a 40% rise in sales in that state three months after the training programme.

According to the sales manager, the training programme was successful, so the salesmen who work in other states should also go through the same training. It is clear that the sales manager is implying that the training programme has increased sales by 40%. That is to say, the training programme is the cause why sales have increased.

Sales training is referred to as a causal variable by the sales manager, while an increase in sales is referred to as an effect variable. The validity of this assertion can now be questioned. Has there truly been a rise in sales as a result of the sales training? The probable response is that we are unable to definitively determine if the sales rise was brought on by the sales training. There might have been additional elements contributing to the higher sales. Increased product penetration in the distribution channel, a strike at a competitor's factory, weather conditions, a price decrease for the air conditioners, etc. could all have contributed to the increase in sales. Therefore, it is very essential that the sales manager should know that conditions under which proper causal statements can be made. One should make sure that the following three requirements are satisfied before drawing any conclusions about causation:

I. We need very strong evidence to say there is a strong association between an action (a causal variable) and the ultimate outcome (an effect variable).

II. The other condition for the causal relationship is that the action (the causal factor) must precede the observed outcome (effect variable).

III. We must have strong evidence to say that there were no other possible factors (causal factors) that could have resulted in the observed outcome

The first condition is concomitant variation. The degree to which a cause (X) and an effect (Y) occur simultaneously or vary is known as concomitant variation. In our case, a sales increase and a sales training programme would need to happen simultaneously. Concomitant variance between the variables in question is a necessary requirement for establishing causality. It may be important to note that a significant correlation between two variables does not necessarily prove a causality relationship. It is quite likely that the strong association between two variables may be the result of random variation or that both variables may be influenced by an extraneous variable.

The second prerequisite for a causal relationship is that the causal factor (activity) must take place either before or at the same time as the effect factor (outcome). However, the fact that the action precedes the outcome does not establish a causal relationship. It might be a mere coincidence that sales training took place prior to an increase in sales of the air conditioners. Additionally, there is a chance that sales training and an increase in air conditioner sales are strongly associated. However, this does not prove that there is a causal relationship.

The third condition for inferring causation is the absence of other possible causal factors. This means that all other possible factor influencing the outcome (in our case increase in sales of air conditioners) are either absent or are kept constant. It is only then we could say logically that the sales training has resulted in increase of sales of air conditioners. In reality, it is impossible to find the absence of other factors or to hold some factors constant. For example, we know that the sales of air conditioners is influenced by weather conditions. Is it possible to keep weather conditions constant? Or can we be sure that the competitor would not change the price? The obvious answer is "No".

In a case where the outcome is completely determined by only one causal factor, we can say that causal factor is the deterministic cause of the outcome. That is, in this case, the "cause" factor is both necessary and sufficient for the outcome to occur. However, in a situation where the outcome is influenced by a host of causal factors, any one of the causal factors is the probable cause of the outcome. That is to say, it is a necessary but not sufficient condition for the occurrence of the outcome.

There are three possible ways to control the influence of extraneous variables. First, we can exert physical control over the extraneous variable. For example, a company trying to study the impact of two different packaging of a product on sales may control an extraneous variable like price by keeping it constant for both packaging containing the same amount and quality of goods. The second way to control the effect of extraneous variables if physical control is not possible is to randomize the assignment of treatments to test units. The third way to control the extraneous variables is through the use of experimental designs, the discussion of which will follow in the subsequent sections. If the control of the extraneous variable on the dependent variable is not possible by any one of the methods, we say that the experiment is confounded, and such an extraneous variable is called a confounding variable.

Activity 1

Suppose the manufacturer of a particular brand of washing machine decreased the price of his sets by 5%. It was observed that there was an increase in sales during the succeeding four months as compared to what the company had prior to the price reduction. Has the price reduction increased sales? Justify your answer.

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4.4 EXPLORATORY RESEARCH DESIGN

The main purpose of executing exploratory research is when the researcher wants some early insights into a problem to get a better idea about the scope and context of the problem. Further, exploratory research can help the researcher understand the research problem more clearly. In other words, when there is a need to define the problem precisely, identify specific possible solutions to a case problem, or secure more theoretical insights about the problem, before going for the main study, exploratory research may be deployed. At the level of exploratory research, the information that is required has a very loose definition. Further, the process of conducting exploratory research is largely unstructured and has sufficient flexibility. Typical exploratory research may involve group discussions and depth interviews, with the sample for the studies being small and generally acquired conveniently. The focus of the research is more information per respondent than collecting data across an array of respondents. Consequently, the data collected is qualitative and analysed using non-statistical techniques. Given the lack of representativeness of the sample in exploratory research, the findings of the research are considered to be tentative and further research may be required to confirm the findings on a much larger sample, referred to as conclusive research. However, not always is exploratory research followed by other research as in some studies, the depth of information and insights are more important than the size of the sample or its representativeness. However, for such research, proper care should be taken before deploying any strategic way forwards provided by such a study.

The purpose of exploratory research is generally to examine and evoke a research problem based on the provided situation and to generate insights for further examination by a researcher.

All marketing research studies should ideally begin with exploratory research, as this helps in providing a sharper focus on the situation and a clearer definition of the problem at hand. The exploratory research design emphasises the finding of ideas and potential insights that may help in identifying areas for more in-depth study. As the name implies, it entails obtaining a sense of the situation. For example, before truly defining the product concept, a food product producer who wants to introduce a breakfast cereal may wish to know the desirable attributes of such a product. The main objective of the exploratory research is to fine-tune the broad problem into a specific problem statement and generate possible hypotheses. It therefore gives useful direction for further research.

The following are the primary objectives of exploratory research:

1. Define a research problem.
2. Identify various courses of action.
3. Create propositions and hypotheses.
4. Identify primary variables and their inter-relationships
5. Developing an approach for the problem.
6. Suggesting priorities for future research in the domain.

Generally, exploratory research is useful when the researcher lacks a clear understanding of the problem before the commencement of the research project or does not have the right variables to be included in the research instrument required for large-scale data collection. An exploratory possesses flexibility as the formal research methods mandated in a particular type of research situation is not pre-specified in this case. For example, for a research situation at hand, the researcher is free to deploy primary methods like focus groups, depth interviews, or projective techniques. Alternatively, the researcher can choose to conduct desk research to review the existing research in that domain. All methods in the exploratory research domain do not involve structured research instruments, like survey questionnaires, and large samples procured through scientific sampling methods. Rather, as researchers conduct exploratory research, they progressively get exposed to novel insights which then direct the researcher into new directions as the research progresses, may redirect their exploration in new directions. The researcher will keep chasing a new direction until all the insights in that direction are completely covered and/or a new direction is discovered. Hence, in exploratory research, the key focus of investigation keeps evolving as the research progresses and as and when new insights are evoked. Overall, conducting effective exploratory research requires a good amount of training and ingenuity on the part of the researcher. Typical exploratory research may involve the following methods:

1. Expert interviews
2. Pilot surveys
3. Qualitative analysis of case studies
4. Qualitative analysis of secondary data/literature review
5. Qualitative consumer research (focus groups, depth interviews, projective techniques)

Survey of experienced individuals

Many marketing problems can be solved by speaking with people who have expertise and ideas about the research topic. These individuals could be top executives, sales managers, salesmen, and channel members who handle the product or related products and consumers or potential consumers. The information collection exercise does not involve a scientifically designed survey; it is merely an attempt to gather all possible information about the subject of research from people who have specific knowledge about it. The success of this type of experience survey depends upon the freedom of response given to the respondent as well as upon the expertise and communication ability of the people questioned. However, at this stage, since the researcher also has very limited experience with the research problem, he may not be able to elicit very valuable responses from the individuals.

Survey of existing literature

Published literature presents a very economical source of study for the purpose of hypothesis generation and problem definition. A large variety of published and unpublished data is available through books and journals, newspapers and periodicals, government publications, individual research projects, and data collected by trade associations. A lot of data is also generated internally in the company, and some of it could be relevant to given problem situations. You will read more about types of secondary data and their use in marketing research in Unit 5 on Data Collection. While reviewing existing literature may not provide answers to the research problem, it can certainly help to guide the research process.

Analysis of selected cases

This method involves the selection of a few extreme examples reflecting the problem situation and a thorough analysis of the same. In certain cases, this sort of study may help in identifying the possible relationships that exist between the variables in a given marketing problem situation. The emphasis in this type of study is on understanding the research subject as a whole. For example, if a company is interested in finding out the reasons for the wide variation in the sales productivity of its salesmen, as an exploratory study it could thoroughly analyse the cases of some of its best and some of its worst salesmen. This exercise may aid in the identification of potential relationships between demographic and/or personality variables that may influence variation in sales productivity. The relationships, their extent, and their direction can then be measured using conclusive research designs.

Activity 2

Examine three marketing research projects carried out in your organisation or any other organisation with which you are familiar:

1. What sort of preliminary research preceded these projects?

(A) ....

(B) ....

C) ....

2. What was the type of exploratory design used for this preliminary research?

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4.5 DESCRIPTIVE RESEARCH DESIGN

This type of design is primarily used in preliminary studies to make it easier to describe population parameters and draw conclusions about how two or more variables interact with one another. The nature of a description or inference may be quantitative or qualitative. Descriptive designs mainly aim to establish a relationship between the elements while describing the event being studied. The information gathered could be related to the situational, behavioural, or demographic characteristics of the respondents under research. For example, a descriptive research methodology can be appropriate to assess a training programme, retail situation, or the various attributes of effective salespeople.

The design could be used to research consumer behaviour in response to a new sales promotion programme. However, it does not establish the extent of association between the different variables, i.e., the income and age of people, as associated with response to the sales promotion. Descriptive design, however, can be used to make inferences about possible relationships between variables. Very often, the decision makers choose to accept descriptive data, which would permit inferences about causality between variables. They may not want to or be able to afford experimental studies in terms of the time involved, and as such, descriptive designs may at times be used for conclusive studies as well.

Descriptive designs are very common, perhaps the most common type of research design. In short, descriptive research can be used for the following purposes:

a) To describe the characteristics of certain groups of interest to the marketer, e.g., users of the product, potential users, non-users, possible receivers of promotional communication by the company, and so on.

b) To estimate the proportion of people in a given population who behave in a certain way, for example, the proportion of consumers who are prone to deals.

c) To make specific predictions for specified future periods.

d) To make inferences about whether certain variables, such as income and shopping location preference, are related.

So, the primary objective of descriptive research is to create a description of something which is usually the characteristics of the market variables or the consumers. Any study can follow the protocols of descriptive research for the reasons to propose a description of the characteristics of groups comprising of consumers, salespersons, firms, or markets. An example can be developing a profile of the heavy shoppers of a prominent departmental store.

1. To describe the percentage or proportion of units in a specific consumer market showcasing a specific behaviour or sharing a common individual characteristic. An example of such a descriptive study will be to calculate the percentage of shoppers at a departmental store who shop for more than one day in a week to be characterized as heavy shoppers.

2. To examine the consumer perception of the characteristics of a product/ service characteristics. An example of such a descriptive study will be the perception of Reliance Smart Bazaar being 'conveniently located' amongst the heavy shoppers of the place.

3. To examine the relationships and associations amongst the marketing variables of interest. For example in the context of a study for a department store, a descriptive study can examine the association between shopping intentions for groceries and those for ice cream by heavy shoppers.

4. To use the proposed model to make specific predictions. An example can be to predict the sales of groceries at a departmental store based on the predicted advertising, store locations, and average grocery prices.

In descriptive research, the researcher needs to have a-priori knowledge about the various ingredients of the research problem at hand, some of which may be acquired through exploratory research conducted before the commencement of the descriptive research. In other words, the relevant concerns of research should be identified through exploratory research before the descriptive research is planned and executed by a quantitative researcher. A prominent requirement of descriptive research, that separates it from exploratory research, is that it mandates a-priori formulation of hypotheses, which can be empirically tested by the data obtained from the study. Such a requirement is not of concern in exploratory research, which is deployed to formulate these hypotheses. Thus, for descriptive research to be executed, the required information to be captured through the research instrument needs to be defined upfront. Thus, compared to exploratory research, descriptive research is much more planned and structured in the format in which enquiry is done with the respondents. The sample size requirements from descriptive research are also high to ensure that the findings of the research are close approximations of the population characteristics.

A proper research design for descriptive research needs to clearly articulate the source of the information needed (population definition), the place to capture the information, the information itself expressed as structured questions in the research instrument, the format for collecting data from the respondents, and the structured analysis to be done on the data to validate the hypotheses. A descriptive research design mandates the who, what, when, where, why, and way of the research to be conducted. These are described below:

1. Who—who is the consumer of a particular brand? He/she can be anyone who has visited the store of the brand and bought something, anyone who visits the brand's store once a month at least, and the individual is the one responsible for the purchase of the category from the brand in his/her household.

2. What—what is the information to be captured from the respondents? Such information could include the frequency with which the respondent visits a particular brand's store, the various choice criteria an individual uses before visiting a store, various factors that help describe the consumer (demographics, psychographics, attitudes, lifestyles, interests, and opinions), and any information that can help validate the hypotheses proposed in the study.

3. When—at what time should the information be captured from the respondents? These could be a specific time of the year, month, day, before visiting a store, while at the store, exiting the store, or sometimes after visiting the store.

4. Where—the location at which the information has to be captured from the respondent. These could be at the brand's store, outside the store, in the mall, at the respondent's home/office, or in a public space.

5. Why—the reason why the specific information is to be captured from the respondent. The reasons could be to examine the perception of the consumer about a brand and its stores, evaluate the consumer loyalty towards the store, perception of the product categories available at

a brand store, develop an advertisement campaign for the brand, determine the location of new stores of the brand, and management of prices of categories available at the store.

6. Way—the protocol to be used for collecting the required information from the respondents. This could be accomplished through observation of respondents, personal face-to-face interviews, telephonic conversations, questionnaires sent through e-mail, or online interviews.

Cross-sectional design

The most frequently deployed research design within the ambit of descriptive research is the cross-sectional research design. This design involves data collection from the members of the sample only once. In other words, a respondent who is part of the study gets to fill out the questionnaire only once. Cross-sectional designs can be further divided into single cross-sectional or multiple cross-sectional designs. In a single cross-sectional design, there is only one sample that is drawn from the population for the purpose of the study and the required information is collected from the members of this sample only once. Such a research design is also referred to as a sample survey research design. In the case of multiple cross-sectional designs, more than one sample is drawn from the population and data from each of these samples is collected only once. Generally, in this research design, there is a temporal gap in the periods when the data is collected. The multiple cross-sectional designs allow the researcher to compare across the two samples at aggregate levels, but not at the individual level. This is because different individuals are involved in filling the questionnaire at two or more points in time and hence, the across-individual comparison is not possible.

The aim of the cross-sectional study is to take a one-time survey of the situation or phenomena that interests the decision-maker. A depiction of the condition at a certain period can be obtained using cross sectional designs. A cross-sectional design might be used to collect data if, for example, the marketer is interested in knowing how the firm stands in the market relative to its competitors or how many viewers overall remember the company's advertisement. This type of design gives a "cross section" of the issue since the information gathered is based on replies from a sample that includes a wide range of sources. In a variety of marketing contexts, cross-sectional data is helpful. Consider the following examples:

- A company of instant food has given display racks to its retailers so that they can prominently showcase their product. Before starting a second round of distribution of these racks, the company wants to ascertain how many merchants and what kinds of retail outlets are actually utilising the racks.

- A company that manufactures electronics goods wants to assess how well its consumers respond to its high-quality after-sales service.

None of the examples given above seeks to establish causal relationships. Cross-sectional studies, on the other hand, may uncover relationships that can be tested conclusively using experimental designs.

Longitudinal design

Unlike cross-sectional design, in a longitudinal design, a specific sample is drawn from the population and the questionnaire is administered to this sample at two or more points in time. In longitudinal research, compared to multiple cross-sectional designs, the sample remains in the different phases of data collection and the same individual gets to fill out the questionnaire two or more times. In other words, longitudinal not only provides a snapshot at a point in time for the sample but also evokes any trends that emerge for various variables across time. This is because the individual filling the questionnaire remains the same and the change in the response given to any variable can be compared across time. An example of cross-sectional design would be to evaluate a brand's perceptions by consumers at a point in time. On the other hand, longitudinal research would be able to measure the change in perception of the brand from one point in time to another.

Another term used for longitudinal design is **panel design** since the panel consists of individuals who have registered with a research firm and have agreed to provide responses to questionnaires over a period of time for any study where they qualify as respondents. The panel members are generally compensated for their participation, and such remuneration can range from gifts, discount coupons, or cash/credit.

A fixed panel or sample of respondents is used in the panel design to collect data continuously or infrequently. Retail establishments, dealers, customers, or just individuals may make up the elements of these panels. Adjustments are performed to take representativeness and dropouts into account, and a panel is anticipated to remain stable over time. A thorough understanding of the panel members' response patterns can be obtained by ongoing measurements of variables related to them, which may also reveal indicators of future behaviour. Repeated measurements of the same variables are used in the longitudinal analysis to allow for a number of conclusions to be made about how the panel's components behave. For example, information on attitudes toward a proposed packaging change may be requested at one point, while responder reaction to advertising copy may be the emphasis at another.

Focus Groups

The focus group, which has an informal approach, is gathering a group of people to engage in a extensive, free-flowing discussion about their experiences or viewpoints.

The standard procedure is to record the conversations on tape and later analyse them. The discussion in the focus group is led and guided by a moderator, who has a fairly standard way of establishing a relaxed and congenial atmosphere, initiating the discussion, and keeping it on the desired topic. The goal of the focus group design is exploratory, with the goal of gathering respondent opinions and experiences on a specific subject. The objectives of the focus group study are clearly specified, but there is no structure to the discussion pattern. In fact, a distinct effort is made to keep the discussion free-flowing. The success of the focus group design depends upon a well-trained and disciplined moderator who can keep the discussion on track and elicit responses from all the participants in the group. Consider the following example:

An established toilet soap manufacturer wishes to alter his advertising theme for a prestigious brand of soap so as to strengthen its image as a luxury soap by focusing on group of young women, including college students, housewives, and some executives. A free-Wheeling discussion about expectations from toiletries in general and bath soap in in particular was generated and recorded in order to provide useful ideas and insights for the creative strategy to be used for the advertising theme.

Focus groups serve as excellent tools for preliminary research as they reveal problems and opportunities that emanate from the free and frank discussions by the consumers about their problems, fancies, expectations, satisfactions, and dissatisfactions. They are easy to organise and interesting to analyse, and they become very good sources of ideas for campaigns, product improvement or repositioning, the desirability of deals, dealer selection, and so on. The danger associated with focus groups lies in the possibility that once analysed, the focus group data may be used for conclusive purposes, to draw conclusions, or to determine how people behave. The size of the focus group being very small, it is dangerous to draw conclusions about consumers' behaviour only on the basis of the recorded and duly analysed discussions.

4.6 FACTORS INFLUENCING EXPERIMENTS VALIDITY

Before taking up the discussion on Quasi-Experimental Designs and Experimental Designs, it would be worthwhile to understand the factors that influence an experiment's validity. There are two important concepts of validity that are related to experimentation: internal validity and external validity.

Internal validity: It means that no other plausible cause of the observed results should exist except those tested. This is the basic minimum that should be present in an experiment before any conclusion about treatments can be made. The presence of other extraneous factors or the absence of internal validity makes the experiment confounded.

External validity: It relates to the question of "generalization" of the results obtained from the experiments. We'd like to know if the experiment's findings can be extrapolated to what population, what geographical areas, and so on.

Although, we would want an experiment to satisfy both internal and external validity, it is not uncommon to find experiments that meet all the internal validity requirements but are invalid externally. This could possibly happen because of the following reasons:

(i) The environment at the time of the test may be different from the environment of the real world where the decisions are taken.

(ii) Results obtained in a six-week test may not hold up in an application of, say, twelve months.

(iii) The population used for experimentation in the test may not be similar to the population where the results of the experiment could be applied.

iv) Treatments in the test may be different from those used in the real world.

It is desirable to understand the sources of errors that are responsible for reducing experiment's internal and external validity. This will help the researcher guide the choice of design to use.

There are several extraneous elements that can affect an experiment's validity. They are described below:

History: History deals with the occurrence of specific, nonrecurring events that are external to the experiment but occur at the same time as the experiment. Suppose we measure the sales of a territory before and after salesmen were trained. The difference in sales may indicate the treatment effect (sales training). However, an increase in sales cannot be attributed to sales training alone. An improvement in business conditions and sales training might have resulted in the increase in sales. Therefore, the longer the time period involved between observations, the greater is the chance that history will confound an experiment of this type and hence significantly affect the results.

Maturation: Maturation is similar to history, except that it is concerned with the changes that occur with the passage of time in the people involved in the design. In our sales training examples, an increase in sales may be due to the fact that the salesmen have become more experienced with the passage of time and, hence, know their customers better. It may be noted that it is not only people who change over time but also stores, geographic regions, organizations, etc. The duration of the time period has a direct impact on the results of the maturation effect.

Testing: In the process of an experiment, respondents may know that their behaviour is being observed or the results are being measured. This would sensitize and bias respondents. In our example of training salespeople, if the salespersons know that they are being trained to know how effective the training is, they might respond differently than what they would have done otherwise. For example, if sales in a territory are recorded prior to salesperson training and another sales observation is made after training, and the respondents are aware of it, they are likely to become sensitised and behave differently.

As another example, consider a case where the respondents have filled out a pre-treatment questionnaire. The respondents are likely to respond differently if they are asked to fill out the same questionnaire after they are exposed to the treatment. This is because they have now become "experts" with that questionnaire.

Instrumentation: This refers to the effect caused by changes in the measuring instrument or process during the experiment. If the difference in total rupee sales volume per territory between "before" and "after" training is used as the yardstick to measure the effectiveness of sales training, a price change during this time interval could make a substantial difference.

This is one example of changing instruments. The experience of the investigators, a change in the mood of the investigators, and a change in the investigators themselves are some other examples of factors that may affect the measurement and hence the interpretation of the results.

Selection Bias: The selection bias occurs when test units are selected in such a way as to not be representative of the population. It refers to the assignment of test units to treatment groups in such a way that the groups differ before being subjected to treatments. If groups are selected based on the judgement of the researcher, there is always the possibility of selection bias. The selection procedure should be random, as this will result in a measurable random variation. However, if a non-random procedure is adopted, the results will be affected by a non-measurable error.

Test Unit Mortality: It is quite possible that some of the test units might drop out of the experiment while it is in progress. Referring to the sales training example, some of the salespeople who are undergoing training might quit before the training is over. There is no way of knowing whether those who were not improving left the training. Also, it is not possible to determine whether those who left the training would produce the same results as those who stayed until the completion of the training programme. In short, the interpretation of the results will become a difficult task.

Activity 4

Explain with the help of suitable examples the factors that influence experiment's validity.

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4.7 QUASI-EXPERIMENTAL DESIGNS

In order to facilitate the discussion on Quasi-experimental designs and Experimental designs (to be discussed in Section 4.8), the following notational system will be used:

i) X represents the exposure of a test group to an experimental treatment whose effect is to be observed or measured.

ii) O refers to the measurement or observation of the dependent variable on the test units. Suppose more than one observation or measurement is taken, the subscripts O_1, O_2, O_3, etc., will be used to indicate temporal order.

iii) R symbolizes that individuals have been selected at random to separate treatment groups or groups have been allocated at random to separate treatments.

iv) Movement from left to right indicates a time sequence of events.

v) All notations in one row indicate that subjects belong to that specific treatment group.

vi) The vertical arrangements of the notations mean that the events occur simultaneously.

To make the national scheme clear, let us consider the following example:

Example 1

The symbols O_1 X O_2 represents that the measurement were taken on a group both prior to (O_1) and after (O_2) of receiving the treatment (X).

Example 2

The symbols

R   O1 X O2

R   X O3

Indicate that two groups of individuals were assigned at random to two different treatment groups at the same time. The groups received the same treatment (X). One group received both a pre-test (O1) and post-test measurement (O2). The second received the post-test measurement (O3) at the same time as O2.

Now having understood the notation scheme, we begin our discussion on Quasi experimental design.

Quasi – experimental Designs

In these designs, the researcher has control over the data collection procedures but lacks control over the scheduling of the treatments and also lacks the ability to randomize test units' exposure to treatments. There are various designs that fall under the category of quasi-experimental designs. Some of these will be discussed here. These designs have an inherent weakness because their internal validity is questionable. They lack the control attributes of truly experimental designs. The following quasi-experimental designs will be discussed in this section.

1) After-Only without Control Group

2) Before-After without Control Group

3) The Static-Group Comparison

4) Longitudinal Designs (Time Series Designs)

1) **After-Only without Control Group**: This design is also called a "One-shot case study." The design may be presented symbolically as follows:

   X O

   In this design, test units are not selected at random. The test units are either self-selected or arbitrarily selected by the experimenter. A single group of test units are first exposed to treatment (X), and then measurements are taken on the dependent variable afterwards (O).

   For example, a sales manager may request that the members of his sales force volunteer for participation in the sales training programme and their sales performance is measured after the training programme is completed.

It is not possible to draw meaningful conclusions from such a design. This is because we do not have any observations (measurements) on the sales performance of the salespeople prior to their undergoing training.

The level of O is the result of various uncontrolled factors. The effects of history, maturation, selection, and mortality are substantial and non-measurable, making the design internally invalid.

2) **Before-After without Control Group**: This design is also termed a "One group pre-test-post-test design." This design is superior to the design" After-Only without Control Group." It provides a measurement of what existed prior to the test group being subjected to the treatment. This design is presented symbolically as follows.

O1 X O2

Referring to the sales training example discussed in "After-only without Control Group," a pre-test measurement of sales performance has been added to the design discussed earlier. Now the question is whether the difference in sales performance as measured by O2 - O1 can be attributed to the sales training program. A number of extraneous factors could be responsible for this difference (O2 - O1), rendering this design ineffective and disqualifying it for internal validity. Some of these factors are listed below:

(i) The test units had volunteered for the sales training programme (selection bias).

(ii) The economic environment might have improved during that period (history).

(iii) The salespersons might have gained more experience during the training period, which would have resulted in improved sales performance (maturation).

(iv) The pre-test measurement might have affected performance (testing).

(v) Prices might have changed during that period (instrumentation)

(vi) Some test units might have left during the period of training (mortality).

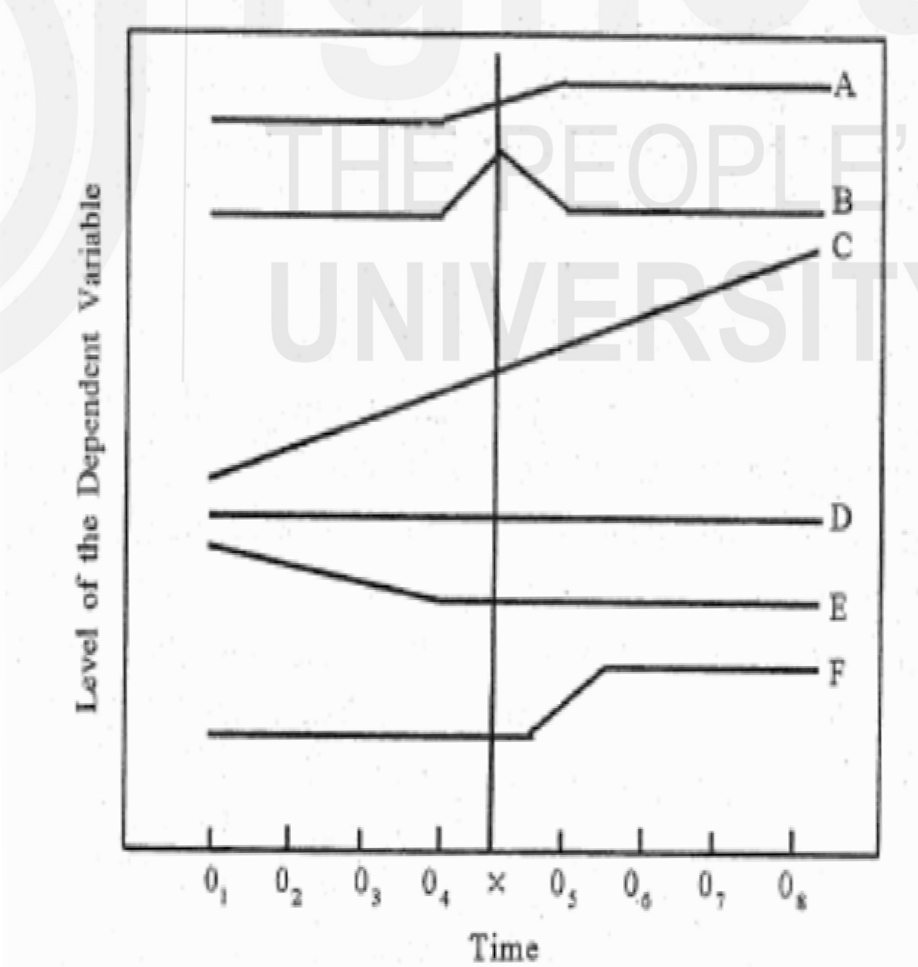
3) **The Static-Group Comparison Design**: In this design, we use two groups of subjects. Group 1 is exposed to the treatment, whereas Group 2 has not been exposed to it. Group 2 is called a control group and is used as the baseline for comparison since it has not been exposed to the treatment. In marketing, the control group treatment is called the current level of marketing activity. The design may be diagrammed symbolically as follows:

Group 1 (Experimental Group):	X1	O1
Group 2 (Control Group):	X2	O2

Here X2 represents the baseline marketing programme (the existing programme) which is to be compared with the new programme (the treatment), X1. For example, X1 could be a new sales training programme which is to be compared with the existing programme X2 (may be called control group treatment). The experiment results O1-O2 could be attributed to the two-group selection procedure. The test units were not randomly selected. So, the result may be attributed to the pretest difference caused by selection procedures. Test unit mortality could also be another probable cause. These two reasons put together could make this design invalid.

5. **Longitudinal Designs (Time-Series Design):** This is an extension of the one-group pre-test-post-test design. Here we take periodic measurements for some test units. The treatment is introduced, and the measurements on the test units are again taken at various points in time. For example, a company may be interested in studying the market share of one of its products. It takes measurements of the market share at periodic intervals. Then a new advertising campaign is introduced, and periodic measurements of the market share are taken again. In order to explain this design, let us consider the figure 4.1 and explain the results of this type of design.

Figure 4.1: Possible Results from Time- Series experiments



At the outset, this design may appear to be similar to one group's pre-test- post- test design. However, there is a difference between the two designs. In the pre-test - post-test design, we would have obtained only two measurements, namely 04 and 05. Here in this design, we have in fact taken many pre-test-post-test observations so that there is more control over extraneous variables. In the pre-test-post-test design, where we use only two observations (04 and 05), the analysis of various situations is as follows:

(a) The results for situations A, B, and C are identical, namely that the market share has increased.

(b) For situations D, E, and F, the conclusion is that the new advertising campaign (X) has no impact on market share as it has remained constant.

The conclusions arrived at from the various situations of the longitudinal designs are as follows:

i) In case A, the campaign had a short-term positive effect, after which the market share remained stable.

ii) In situation B, it is observed that the rise in market share was temporary. The new campaign had a short-run positive effect. We note that the market share reverted to the level existing before the application of the treatment.

iii) In situation C, we observe that the changes that occur after the application of the treatment are a continuation of a trend (they also existed prior to the application of the treatment). Therefore, it is quite likely that the new advertising campaign had no effect on the market share.

iv) In situation D, the conclusion would be the same as that obtained for situation C.

v) In situation E, it appears that the new advertising campaign seems to have stopped the decline in market share.

vi) In situation F, the treatment (the new advertising campaign) had a delayed effect and therefore required a longer period to appear.

Because of the multiple measurements in this design, the effect of maturation, testing, and instrumentation can be ruled out. Further, if the test units are selected randomly and measures are taken to prevent panel members (test units) from dropping out, one could at least reduce the selection and test unit mortality factors. However, the main weakness of this design is that it is not possible to control history. However, the researcher could maintain a record of various events like unusual economic activity, economic changes, etc., and if no changes are observed, one could reasonably conclude that treatment has had an effect.

Activity 5

Describe a few specific situations from the functional area of marketing where you would recommend each of the following designs as being best suited. Justify your answer.

a) After-Only Without Control Group

b) The Static-Group comparison

c) Longitudinal Designs.

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4.8 EXPERIMENTAL DESIGN

True experimental designs provide a stronger and more reliable basis for the existence of casual relationships between variables. Here, the researcher is able to eliminate all extraneous variables from the experimental and control groups through the use of a random selection procedure. One of the advantages of using a random selection procedure is that we can use inferential statistical techniques for the analysis of experimental results. One such technique is the analysis of variance. The following experimental designs will be discussed in this section.

1) After-only with One Control Group
2) Before-After with One Control Group
3) The Solomon Four Group Design
4) Completely Randomised Design
5) Randomized Block Design
6) Latin Square Design

1) **After-Only with One Control Group**:- Here, two groups are experimental and one control group are set up by randomisation procedure. The design requires only one treatment and post test measurement of both of experimental and the control group. The design may be written symbolically as follows:

Experimental Group	R	X	O1
Control Group	R		O2

It may be noted that no pre-test measurements are taken in any of the two groups. The difference between the post-test measurement of the two groups is taken as a measure of treatment effect. Here 0, measurement is composed of treatment effect and all the other extraneous variables and 0, is composed of extraneous factors only. Therefore:

O_1-O_2 = (Treatment effect + Extraneous factors) - (Extraneous factors) = Treatment effect.

In this case, the lack of pre-test measurement provides control over the testing and instrument effect. Also, the interactive testing effect cannot occur. We, however, assume in this design that the pretest measurement of the characteristics under study is the same in both groups. It is also assumed that the test-unit mortality affects each group equally. These assumptions can be justified if large, randomly selected samples are used. This design is very widely used in marketing.

2) **Before-After with One Control Group:** This design may be presented symbolically as:

Experimental Groups	R	O_1	X_1	O_2
Control Groups	R	O_3		O_4

This is also known as a pre-post -test design. In this design, two groups—one "experiment" and one "control"—are set up by a randomization process. The extraneous variables operate equally in the experimental and control groups. Pre-test measurements are taken in both groups. However, it is the experimental group that is exposed to treatment. Post-test measurements are taken for both groups.

Because the extraneous factors affect both groups equally, the only difference between the two groups is due to the experimental group's treatment.

Let A = Change in the measurements of the experimental group

$$O_2 - O_1$$

Let B = Change in the measurements of the control group

$$O_4 - O_3$$

Therefore, treatment effect - A-B = $(O_2 - O_1)$-$(O_4 - O_3)$

The main problem with this design is that it is not possible to remove the sensitizing testing effect from the measurement effect. The estimate of the treatment effect given above also includes the sensitizing testing effect. The sensitizing test effects occur in the experimental group and not in the control group. The individuals in the experiment group get affected because of the pre-test measurements and may become very cautious and behave differently after being exposed to the treatment. As a result, the sensitizing testing effect will be included in the measure A = $O_2 - O_1$. This is missing in B = $O_4 - O_3$ because the control group is not exposed to the treatment. Therefore, this shows that the treatment effects as given by A-B will include sensitizing testing effects, and therefore we cannot generalise the result of our treatment.

3) **The Solomon Four Groups Design**: This design is also called the four-group, six-study design. This test overcomes the limitations of pre-test-post-test control group designs. This design is actually a combination of the "Before and After with One Control Group"

and "After Only with One Control Group" designs. The design is symbolically presented as:

Experimental Groups:	R	O_1	X	O_2
Control Groups:	R	O_3		O_4
Experimental Groups:	R			O_5
Control Groups:	R			O_6

You may note that the experimental group 2 and control group 2 did not receive any pre-test measurements. Because this type of sensitising cannot occur in measurement O_5. The results of group 2 assist us in measuring and eliminating the sensitising testing effect. The results of the difference between various pre-test and post-test measurements give us the following effects:

Experimental Groups	:	$O_2 - O_1$ = treatment effect + extraneous + sensitizing testing effect)	(i)
Control Groups 1	:	$O_4 - O_3$ = extraneous factors	(ii)
Experimental Groups 2	:	$O_5 - O_1$ = (treatment effect + extraneous factors)	(iii)
		$O_5 - O_3$ = (treatment effect + extraneous factors)	(iv)
Contol Groups 2	:	$O_6 - O_1$ = extraneous factors	(v)
		$O_6 - O_3$ = extraneous factors	(vi)

By taking the average of equations (iii) and (iv), we get

$$O_5 - \frac{O_1 + O_3}{2} = \text{Treatment effect + Extraneous effect} \quad \text{(vii)}$$

By taking the average of equations (v) and (vi), we get

$$O_6 - \frac{O_1 + O_3}{2} = \text{Extraneous effect} \quad \text{(viii)}$$

Treatment effect

$$= \left(O_5 - \frac{O_1 + O_3}{2}\right) - \left(O_6 - \frac{O_1 + O_3}{2}\right)$$

$$= O_5 - O_6 = \text{Treatment effect}$$

By substracting equation (vii) from equation (i), we obtain the sensitizing testing effect as:

$$\text{Sensitizing testing effect} = (O_2 - O_1) - \left(O_5 - \frac{O_1 + O_3}{2}\right)$$

Please note that in order to measure the change in experiment group 2, we required an estimate of the pre-test measurement on O_5. We assumed that groups were equal before the experiment because of the random assignments of the subjects to all four of our group 1 and control group 1 respectively. Given the just-mentioned assumption, the pre-test measurement of experimental group 1 and control group 1 could be taken as an estimate of the pre-test measurement of experiment group 2. That is what we used in the equations (iii), (iv), (v), and (vi) listed above.

This design is also referred to as "the ideal control experiment." As you have seen, we are not only able to control all extraneous variables and the sensitizing testing effect but also get their estimates. However, this design is not very commonly used in marketing because of the increase in cost and the time and effort required to conduct the experiment.

The three experimental designs discussed above allow us to manipulate or control an independent variable so as to measure its effect on the dependent variable, thereby enabling us to make proper casual statements. In the sections that follow, we will look at some designs in which we can manipulate or control more than one level of an independent variable on the dependent variable.

4. **Completely Randomised Design:** The test is useful when the investigator is interested in studying the effect of one independent variable on the dependent variable. The independent variable should have a number of categories (a nominal scale). Each category of the independent variable is considered a treatment. Suppose we wish to sell a product and want to know what price should be charged. Here the independent variable is price, and the three different levels of price, namely low, medium, and high, are used as treatments. The test units (shops) are randomly assigned to three different treatments (price levels). Assume our dependent variable is; we could always compare the average sales level for each treaent to determine which treatment (price level) was the best.

   The limitation of this design is that it does not take into account the effect of various extraneous variables on the dependent variable. In the example that we have considered, the possible extraneous factors would be the competitor's price, the size of the shops, the price of the substitute product for the product in question, and so on. It is assumed in this design that the extraneous factors have the same impact on all the test units.

5. **Randomized Block Design:** The randomized block design is an extension of the completely randomized design. In the completely randomized design, it was assumed that all the extraneous factors would have the same impact on all the test units. This may not be true.

   In the examples used in the completely randomized design, we had three price levels tested in various shops. We did not take into consideration the fact that the differences in the results could be due to the differences in the sizes of the shops. One could classify various shops on the basis of their size as follows:

   Small-sized shops

   Medium-sized shops

   Large-size shops

   In the randomized block design, it is possible to separate out the effect of one extraneous factor from the results, thereby providing a clearer picture of the impact of treatment on sales.

   In this experiment, we may, for example, randomly assign nine small size shops to the three price levels in such a way that there are three shops for each price level. Similarly, one could assign randomly nine medium-sized shops to the three price levels and nine large-sized shops to the three price levels. Having done this, one could use the analysis of variance technique to separate out the effect of the extraneous variable (size of shops) from the total experimental error and also analyse the effect of the treatment (price level) on sales (the dependent variable).

6. **Latin Square Design:** This design allows the researchers to control and measure the effect of two extraneous variables on the dependent variable. Here the two extraneous variables need to be divided into as many categories as the number of levels of the independent variable (treatment). Table 4.1 illustrates the layout of the Latin Square Design.

Table 4.1: Latin Square for Various Advertising Campaigns

		Packaging			
		I	II	III	IV
Cities	1	A	B	C	D
	2	B	C	D	A
	3	C	D	A	B
	4	D	A	B	C

Suppose a marketing organisation wants to study the impact of four types of advertising campaigns on the sales of a product. The sales could be affected by, say, two extraneous variables like the type of city and the type of packaging. If Latin Square Design is to be used, the cities need to be divided into four types, namely 1, 2, 3, and 4, and four different types of packaging, namely I, II, III, and IV, are required. In Table 4.1, we have identified four categories for the row variable, the type of city, and four categories for the column variable, the type of packaging. The treatment variable is the type of advertising campaign, which is identified as A, B, C, and D. We note that the number of categories of extraneous variables is "the same as the number of treatments" (4).

The treatments (four advertising campaigns) are assigned randomly to the cells of the table in such a way that each treatment appears once and only once in each row and each column. Once this arrangement is made, the analysis of variance technique could be used to separate out and measure the effect of the two extraneous variables in question and also analyse the effect of treatments (four types of advertising campaign) on the dependent variable, namely sales of the product.

This design is very complex to set up. Further, it can be very expensive to execute.

Activity 6

How would you use a "Before-After" design with one control group to measure the effectiveness of new advertising on the sales of a particular brand of shampoo? Also, mention the weakness of this design.

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4.9 SUMMARY

An overview of research designs and how they are used in marketing research is provided in this unit. Following a review of the purpose and advantages of a research design, the discussion concentrates on the four primary types of designs: exploratory, descriptive, quasi-experimental, and experimental. The foundation of this classification is causality, or the relationship between the data and the conclusions. There have also been discussions on the nature of causality and the need to measure it.

The major design categories and their applications have been described. Specific subtypes within the major categories have also been discussed. Under descriptive design, panel design, cross-sectional design, and focus groups were discussed. The quasi-experimental designs discussed include after only without a control group, before-after without a control group, the static group comparison, the longitudinal design, and multiple time series designs.

Under the experimental design categories, the designs covered are after-only with one control group; before-after with one control; Solomen four group design (completely randomized design); randomised block design; and Latin square design. The internal and external validity of the experimental validity of the experiment were also discussed.

4.10 KEY WORDS

Cross-sectional Design: This design involves data collection from the members of the sample only once.

Descriptive Research Design: This type of design is primarily used in preliminary studies to make it easier to describe population parameters and draw conclusions about how two or more variables interact with one another.

Experimental Design: True experimental designs provide a stronger and more reliable basis for the existence of casual relationships between variables. Here, the researcher is able to eliminate all extraneous variables from the experimental and control groups through the use of a random selection procedure

Exploratory Research Design: Exploratory research design is used when the researcher wants some early insights into a problem to get a better idea about the scope and context of the problem.

External validity: It relates to the question of "generalization" of the results obtained from the experiments.

Focus Groups: The focus group, which has an informal approach, is gathering a group of people to engage in a extensive, free-flowing discussion about their experiences or viewpoints.

History: History deals with the occurrence of specific, nonrecurring events that are external to the experiment but occur at the same time as the experiment.

Instrumentation: This refers to the effect caused by changes in the measuring instrument or process during the experiment.

Internal validity: It means that no other plausible cause of the observed results should exist except those tested.

Longitudinal design: Unlike cross-sectional design, in a longitudinal design, a specific sample is drawn from the population and the questionnaire is administered to this sample at two or more points in time.

Maturation: Maturation is similar to history, except that it is concerned with the changes that occur with the passage of time in the people involved in the design.

Quasi – experimental Designs: In these designs, the researcher has control over the data collection procedures but lacks control over the scheduling of the treatments and also lacks the ability to randomize test units' exposure to treatments.

Research Design: The research design is a detailed blueprint for the research study that will be conducted and provides a general outline of the procedures that will be used.

Selection Bias: The selection bias occurs when test units are selected in such a way as to not be representative of the population.

Test Unit Mortality: It is quite possible that some of the test units might drop out of the experiment while it is in progress.

Testing: In the process of an experiment, respondents may know that their behaviour is being observed or the results are being measured. This would sensitize and bias respondents.

4.11 SELF-ASSESSMENT QUESTIONS

1. Distinguish between exploratory and descriptive research design.

2. Explain the various extraneous variables that, if not controlled in an experiment, may contaminate the effect of the independent variable.

3. What are the marketing research situations suitable for

   a) Focus group study

   b) Panel research design

   c) Cross-sectional design

   d) Quasi-experimental design

4. What design would you use to

   a) Test whether the coloured newspaper advertisement is more effective than the existing black and white advertisement?

   b) Know which of the several promotional techniques is most efficient in selling a particular product?

c) Determine whether additional shelf space allocated to a product would increase the sales of that product in the supermarket?

d) Determine the type of customers who would be the first adopters of a newly introduced brand?

Justify your answer.

4. Explain the solomon-four group design. How far does this design succeed in controlling different extraneous variables? Illustrate your answer with the help of a suitable example from marketing.

4.12 PROJECT QUESTION

A leading ice cream manufacturer is planning to launch a new flavour of ice cream aimed at the premium market segment. What type of research design will be appropriate for marketing research? Explain.

4.13 FURTHER READINGS

- Freund, John E. and Frank J. Williams, "Elementary Business Statistics - The Modern kiproach", Prentice Hall International Edition.
- Kinnear, Thomas C. and James R. Tayler, "Marketing Research - An Applied Approach", McGraw-Hill International Edition.
- Luck, David J. and Ronald s. Rubin, "Marketing Research" Prentice-Hall of India Pvt. Ltd.
- Green, Paul E. and Donald S. Tull, "Research for Marketing Decisions" Prentice Hall of India Pvt. Ltd.
- Westfall, Boyd and Stasch, "Marketing Research - Text and Cases" Rechard D. Irwin. Inc.

UNIT 5 DATA COLLECTION: QUANTITATIVE AND QUALITATIVE

Learning Outcomes

After studying this unit, you should be able to:

- explain the necessity and usefulness of data collection.
- distinguish between primary and secondary data.
- identify the sources of secondary and primary data.
- examine the reliability, suitability, and adequacy of secondary data.
- appreciate the advantages and limitations of primary and secondary data
- describe different methods of collecting primary data with their merits and demerits.
- assess the sources of error in primary data collection

Structure

5.1 INTRODUCTION

In unit 4, the research design alternatives available to the researcher were discussed. Whatever the type of research design selected, it is essential to gather precise and trustworthy data in order to meet the research objectives. One of the most important assets that organizations have nowadays is data. The more data you have about your consumers, the better you can comprehend their interests, needs, and wants. Your ability to meet and exceed your customers' expectations, as well as develop appealing messaging and products or services, is improved by this insight.

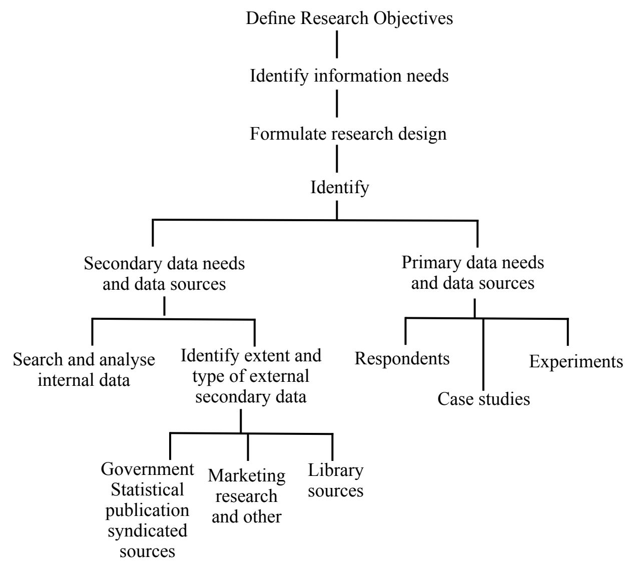
Information technology is increasingly being incorporated into the methods that marketing researchers use to collect data. The development of new data collection techniques that produce valuable information quickly and in real time has been made possible by the World Wide Web and other internet-based technologies. Compared to traditional polls, which require a delay, these technologies deliver information considerably more quickly. Significant changes have occurred and are still occurring in the domain of data collection, where marketing research is dependent on communications. The development of technology and customers' quick adoption of sophisticated personal communication systems have made it possible to collect data in new and more effective ways that are quicker, easier, more secure, and even less expensive. As an illustration, many individuals use smart phones, which include a variety of features like caller ID and conference calling. While these smartphone features make it easier for researchers to connect with respondents to gather data, they also have the opposite effect. The likelihood that someone won't reply is increased by the use of "gatekeepers" like caller ID, answering machines, and call blocking on smart phones.

In this unit, the various data types used for marketing research will be discussed, along with their role in the research process. Additionally, the primary and secondary data sources as well as the basic data collection methods will be discussed. Errors of various forms typically correspond with particular data collection types and methods, whether the data were obtained through observation or communication means. We will also go over some of the most common causes of data collection errors.

5.2 DATA AND THE RESEARCH PROCESS

Once the research objectives have been identified and defined, the next step is to determine the nature and type of information required to accomplish these objectives. The research process is illustrated in Figure 5.1, with special attention paid to the variety of data sources and alternatives available to the researcher. The research objectives define the type and extent of information needed to achieve the research objectives; the data needs are further clarified by the type of research design chosen as well as by the nature of the research, e.g., whether the problem at hand is one of exploratory or preliminary research or causal and conclusive research, and so on.

Figure 5.1: Data and Research Process



All data sources available to the researcher can be classified as either secondary or primary. Secondary data are already published and collected for purposes other than the research problem at hand. Primary data are those that are collected specifically for the purposes of the research problem at hand. In most cases, the initial challenge in the data collection process is to ascertain whether the data required for the research problem has already been generated or collected. Only after evaluating the data that is already available should new data requirements be determined.

Quantitative and Qualitative Approach

Typically, qualitative research is categorized as being unstructured, whereas quantitative research is categorized as being structured. The following factors should determine whether to use a quantitative or qualitative approach:

- The purpose of your research: exploration, confirmation, or quantification
- Application of the results: formulation of policy or comprehension of the procedure

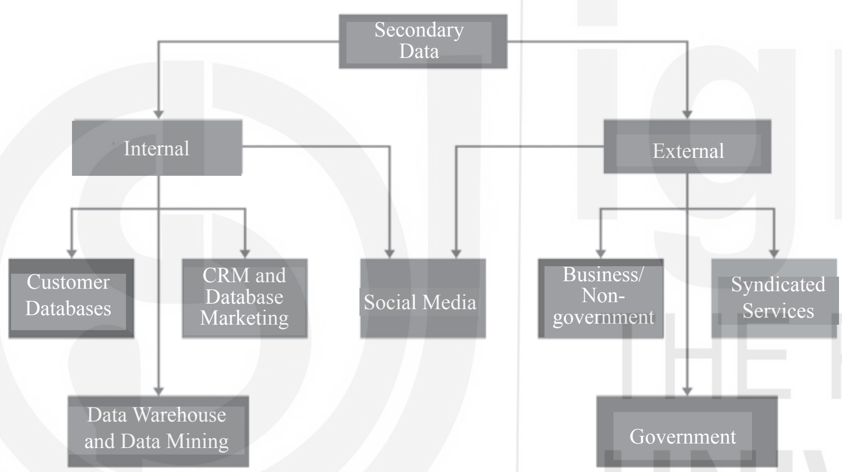
While the research process is mostly the same in both, there are differences between quantitative and qualitative research in terms of the techniques used to gather the data, the methods used to process and analyze the data, and the methods used to communicate the results. Using an unstructured interview or observation as your method of data collection, for instance, is more likely if your research problem lends itself to a qualitative style of inquiry. When analyzing data for qualitative research, you do so by looking for themes and explaining what you learned from your interviews or observations rather than by subjecting your data to statistical techniques.

5.2.1 Types of Data

As you have already noted, there are two general types of data: primary and secondary, classified on the basis of the purpose of collection or source. Primary data are those that are collected specifically for the research situation at hand. Conversely Secondary data has already been published and collected for purposes other than specific research needs. Secondary data can be further classified as internal or external based on the location of the sources. The data originating within or available within the organization as a byproduct of the MIS or the routine reporting system is called internal data of any given marketing research problem, initial data collected for purposes other than that specific problem could be termed internal secondary data, as exhibited in figure 5.2.

Secondary data generated outside the organization is termed "external secondary data" and can be collected from a variety of sources, like government publications, trade association publications, official reports, journals and periodicals, and the publications of marketing research agencies. Additionally, secondary data can be purchased from research agencies, albeit this is a fairly expensive option. Data generated on social media may be internal as well as external source of secondary data.

Figure 5.2: Sources of Secondary Data



Secondary data may also be classified on the basis of whether it is periodic data or ad hoc data. The majority of statistics gathered over set periods of time, such as census data, data from statistical abstracts of trade and other sectors, price indices, and so forth, are characterized by periodic data. On the other hand, ad hoc data, refers to data collected from a particular project report. Identifying the source from which such data can be accessed usually requires an external search.

The original, fresh, and first-timely collected data constitute the primary data and are, thus, considered to be the most authentic. As Figure 5.1 shows, the major sources of primary data are respondents, analogous case studies and experiments. On the other side, secondary data can be obtained from library sources, syndicated sources (which sell the data), marketing research firms, and official government publications. Since published individual research projects are part of the library source, they have not been addressed individually.

Finding out first-hand a customer's attitudes toward specific products or services, evaluating the advertising effectiveness, determining the job satisfaction of employees, and determining the quality of service provided by a service professional are all examples of data collected from primary sources. However, using census data to learn about a population's age and gender distribution, using an organization's records to learn about its operations, and gathering data from articles, journals, magazines, books, and periodicals to learn about historical and other types of information are all considered secondary sources. In summary, primary sources offer first-hand information while secondary sources offer information that has already been reported.

Activity 1

Speak with the person in charge of marketing research in your own company or any other business you are familiar with. Find out what kind of data the organization used to make decisions about:

(a) a new product release, and

(b) the implementation of a new sales incentive programme.

What were the main data sources and the methods utilized to get the data?

	a) Product introduction decision	b) Sales incentive scheme introduction
Type of data	……………………….. ……………………..	………………………… …………………………
Major data sources	……………………….. ………………………	………………………… …………………………
Methods used	……………………… ………………………..	………………………… …………………………

5.3 SECONDARY DATA- NEED AND USAGE

Before we get into the sources of secondary data, consider what kinds of research objectives and information needs might necessitate the use of secondary data. Some examples could be:

- You need to assess the potential market size for Teflon - coated packaging cases in the country.
- As a manufacturer of frost- free refrigerators, you wish to create a nationwide system for setting sales quotas state-wide.
- As a manufacturer of automotive batteries, you wish to determine the potential market for battery replacements and create methods for establishing national, state, and district-wide sales quotas.
- You want to forecast prospective wholesale and retail sales of A4 xerox paper in 2023.
- To locate a new distribution center for the spare parts of your four-wheeler products, you are required to choose a location in western Uttar Pradesh.

- You wish to allocate budget for sales promotion proportionally to all prospective markets in southern India.

Some of the secondary data needed to achieve the aforementioned research objectives include population data, estimates of total frost-free refrigerator sales by state, estimates of the number of refrigerator-owning households, per capita income estimates, statistics on the distribution of income, car registration numbers, the number of households without cars, the value of box shipments by end use, etc. You have learnt about the different types of research designs in the unit 4. The requirement for the use of secondary data sources varies depending on the research design.

When conducting monitoring research, a researcher tries to keep track of a programme or phenomenon in connection to the variables being examined and maintains a watch out for environmental changes that can help or hinder the product programme being investigated. A significant portion of the data used for this kind of monitoring would be created internally, with secondary sources providing the majority of the information. Accounting data on sales, expenses, and profit margins, for instance, could offer important hints about how the product is doing in the specific market. Executives may be made aware of possible problems by observations made in sales force reports regarding consumer behaviour and market trends.

Although new data can be collected in small or informal primary investigations, preliminary research also makes considerable use of secondary data. In this situation, secondary data is used to infer the specifics of a problem or trend as well as changes to its environment. In general, preliminary research does not justify the cost and delay of primary data collecting because it examines the nature and scope of the problem. Usually, descriptive data from internal and external sources can be employed in this initial stage to comprehend and characterize the nature of a research problem.

You will only be able to use secondary data to a limited extent when conducting exploratory research, which is concerned with identifying specific alternatives that should be taken into account when making a decision (for example, when trying to explore the various combinations of the communication mix that could yield the optimal mix). Obviously, previous experiences with the business, its records, or internal data may offer directions for action, as may secondary external data, particularly published content on consumer perceptions and reactions to various instruments in the communication mix, media habits, etc.

Conclusive research relies mostly on primary data collected to accomplish the objectives of the research subject under study, as it is focused with determining the dependence or independence of the variables under inquiry. For instance, you would have to rely primarily on new data created specifically to establish the cause-and-effect relationship between the sales promotion programme and the luggage sold if you were interested in learning whether a particular sales promotion programme has a significant impact on the sales of a new range of moulded luggage suitcase.

However, even in conclusive studies, secondary data may be employed for comparisons. You might prefer to compare the industry trend in terms of marketing spending and sales with the position of your own company or the company's position throughout the state with that of a few of its specific competitors. Additionally, secondary data can be used to frame a primary data collection programme, frame the sample, or calculate the sizes of the sub-samples.

5.4 SOURCES OF SECONDARY DATA

Government Organizations and Official Publications: In terms of secondary macroeconomic statistics, the Union government's body of publications is by far the most important. This data source has been used by marketing researchers to define sales areas and routing plans, as well as to estimate market potential and sales predictions, ascertain distribution penetration, and determine the locations of intermediate and final outlets. Estimates of income and spending habits are an excellent place to start when determining someone's ability to pay for different goods and services. Estimates of literacy levels are useful tools for developing promotional campaigns. Government data have become much more varied and in-depth over time, which has boosted their usefulness for marketing research. The sources listed below are very pertinent for marketing research.

1. **Population Statistics**

   Since 1871–1872, India has collected population statistics every ten years. Census data with regards to the marketing research function is the most significant source of national population statistics and its fundamental characteristics, primarily demographic and economic, in addition to serving as the foundation for the majority of macroeconomic planning estimations. By providing information on the population's number and distribution by age, sex, occupation, and income, it gives factual foundation for the calculation of consumer demand for a range of goods and services. Creating a sampling frame for more trustworthy sample designs can also be done using census data. Estimates of population size, distribution, literacy rates, and consumption trends all play significant roles in determining how to market goods and services, how to communicate effectively, and how to formulate a general marketing strategy.

2. **Statistical Abstract of India**

   The Central Statistical Organization publishes this publication annually, and it contains statistics from various sectors of the Indian economy. State wise, statistics for these economic variables are also furnished periodically.

3. **Economic Survey of India**

   The Economic Survey is a yearly financial report that analyses and provides comprehensive statistical data on all sectors, including industrial, agricultural, and industrial production, employment, prices, and exports, among others. It reviews the economic development in India over the previous fiscal year. It also examines changes in the money supply and foreign exchange reserves, two key elements that have an effect on the Indian economy. The Economic Survey of India is prepared by the Economics Division of the Department of Economic Affairs of the Finance Ministry under the overall guidance of the Chief Economic Adviser (CEA) and is released a day before the Union Budget is presented in Parliament.

4. Monthly Statistics on the Production of Selected Industries

The Central Statistical Organization releases monthly information on output, installed capacity, and stock positions in a few industries to fill the gap between the census conducted and the data published.

5. Trade Statistics

The Directorate General of Commercial Intelligence and Statistics under the Ministry of Commerce and Industry, Government of India, compiles monthly statistics of commercial intelligence and statistics and publishes data on the import and export of goods in terms of their quantity and value, classified as received from or sent to centres of consignment. In addition, this publication provides information on the value of foreign trade, the balance of trade, foreign trade with each country and currency area, foreign trade in groups of commodities with each country and currency area, foreign trade with selected countries, etc. It therefore furnishes a good source of data for assessments of international market trends and potential.

The Reserve Bank of India also publishes statistics on imports and exports based on exchange control data. In addition, official reports, both published and unpublished, from organizations like the National Sample Survey Organization, the Directorate of Economics and Statistics, the Labor Bureau, etc. furnish data on the different aspects of our economy.

Agencies like the Cottage Industry Board, the National Small Scale Industries Corporation, the Tea Board, the Handicraft Board, etc. provide specific information with respect to their own product categories.

6. Publications from foreign governments or international organizations:

The significant publications come from international organizations including the UNO, FAO, WHO, UNESCO, ILO, and the Yearbook of Labor Statistics, which is issued by the Statistical Office of the United Nations (published by the ILO, Geneva). The secondary data offered by such sources is accurate, but in addition to other considerations, one must pay close attention to the units of measurement because they substantially differ from one country to another in terms of weight, currency, and other factors.

7. Other Sources of Secondary Data:

- The **library sources** of marketing data include the whole gamut of publicly circulated material i.e, government documents and reports, books, periodicals, journals, magazines and newspapers, reports prepared by various universities, historical documents, diaries, letters, unpublished biographies, autobiographies, individuals research project reports and trade association publications.

- **Journals of trade, commerce, economics, engineering** etc. published by responsible trade associations, Chambers of Commerce provide secondary data in respect of some important items. Some examples of this kind of publications are "Annual Report of the Chief Inspector of Mines in India" (issued annually by the office of the Chief Inspector of Mines, Dhanbad) and "Indian Textile Bulletin (issued monthly by the Textile Commissioner, Bombay).

- **Various publications of Central, State and local governments**: The important official publications are India-Annual; Monthly Abstract of Statistics (both published by Central Statistical Organisation); Indian Agricultural Statistics (Annual) (Published by Ministry of Food and Agriculture); Index Number of Wholesale Prices in India (Weekly) (Published by Ministry of Commerce and Industry); Reserve Bank of India Bulletin (Monthly) (Published by Reserve Bank of India).

- **Research Agencies and Data Services**

  A number of organizations that gather and sell standardized data have emerged in response to the rising need for marketing data. These organizations, usually referred to as syndicated sources, include marketing research companies and data providers that not only provide standardized data but also carry out specific data collection projects for research. Advertising firms have become reliable providers of data on readership, media usage, attitudinal research, and other communication-related topics in India. These syndicated agencies can provide consumer, retail, wholesale, industrial, advertising evaluation, media, and audience data, among other types of information.

Data Warehouse and Data Mining

Data from several operating systems in the organizations is consolidated into a single, central database known as a data warehouse. The process of analyzing such big databases, known as data mining, calls for specialized knowledge and resources. Large datasets are analyzed using powerful computers, advanced statistical software, and other tools in data mining. The goal is to find hidden patterns in the data. The patterns found can be incredibly helpful for focusing marketing efforts. For instance, analysis of the data showed that couples get additional life insurance right away following the birth of their first child.

Customer Databases

Transferring raw sales data to a computer, such as that found on sales call reports or invoices, is frequently the first stage in the creation of a customer database. Additionally, information on customers is collected from various sources like warranty cards and loyalty programmes. This information is supplemented by demographic and psychographic data about the customers from other secondary sources. These customer databases are kept in a data warehouse.

Activity 2

A) Analyze the demand estimation process for an existing product in your organization or any other organization you are familiar with.

a) What secondary data sources were used?

…………………………………………………………………………..

………………………………………………………………………….

b) How specifically did the study use secondary data?

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B) Repeat the exercise (A) above to estimate the demand for an upcoming new product.

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5.5 ADVANTAGES AND LIMITATIONS OF SECONDARY DATA

5.5.1 Advantages of Secondary Data

The main benefit of secondary data is the resource efficiency it provides, both financially and in terms of time. In primary research, the sample frame, sample size, data collection instruments, printing of the data collection instrument in the case of field studies, editing, tabulating, and analysis of the results are all steps that are both costly and time-consuming. On the other hand, secondary data can be quickly and economically gathered by researchers from published or compiled studies.

Some secondary data providers offer access to information that would not often be available to a single organization, which is another significant benefit. For instance, the census of wholesale and retail businesses may ask them to disclose statistics on sales, costs, and profits that an individual researcher would not have access to. Additionally, since this type of data is gathered in the usual course of events, it is less likely to be biased than data that is gathered specifically for a certain reason.

5.5.2 Limitations of secondary data

Although the economy and having access to comparatively unbiased data collection are important advantages, there are two fundamental problems that must be solved before secondary data can be used effectively. These are the challenges with data **accuracy** and **fitness**.

5.5.2.1 The problem of data accuracy

Once a suitable secondary data source has been identified, it is crucial to assess the accuracy of data in the context of the current research problem. The following steps would help in evaluating the accuracy of the secondary data.

Identify the data source: Secondary data may be obtained from original source or from another secondary source. Data must be gathered from the original sources as much as possible since you may then be able to assess the research methodology that the original researcher used. When secondary sources of secondary data are used, not only are the technique of the original source's details unavailable, but errors occurring at the secondary source could also have an impact on the accuracy of the original data. Additionally, secondary sources frequently fall behind the updated versions and revisions that the primary sources occasionally publish as new information becomes available to them. This necessitates using secondary sources of data with care and only after careful consideration.

Examine the purpose for which data was published:

It is important to use caution when using sources that publish in order to promote the interests of a specific group or for commercial, political, or social objectives. Similar to this, data produced to increase sales, spread a certain propaganda, or advance the opinions of a particular interest group is only useful to the extent that it is carefully assessed in light of the publication's intended audience. Data that is published anonymously, by a group that is on the defensive, in circumstances that suggest a controversy, in a manner that reveals a strained attempt at frankness, or in an effort to refute inferences drawn from other data, are typically suspicious.

Assess the source in terms of the quality of data expected:

It is necessary to determine the organization that collected the data as well as that organization's capacity to obtain the required data. For example, the Internal Revenue department is significantly more likely to produce accurate information regarding income and expenses simply because it has the authority to do so.

The evaluation of the research methodology employed by the source in terms of the research design, sampling plan, data collection methods, and analysis techniques also provides significant insights to the accuracy of secondary data. One should use caution when using the data if the primary source is reluctant to disclose the methodology of research, as this is typically a sign that the techniques employed were insufficient.

5.5.2.2 The data fitness problem

The problem here refers to the suitability or fitness of secondary data for the research problem at hand. Secondary data, which has been compiled for other purposes, is rarely completely relevant to the information needs of the problem at hand. Even when they are relevant, the degree of fitness to a given research problem may be impeded by (a) units of measurement, (b) class definition, and (c) recency.

The variance in **units of measurement** employed for a particular variable is one of the most frequent constraints of secondary data. For example, If you want to base your market estimates on a realistic estimate of disposable income, you might find that, depending on the source, income may have been estimated by individual, household, or tax returns. If more than one of these sources must be used, the information can only be used as a rough approximation due to the different measuring units.

Another characteristic of secondary data is the **variation in class definitions** found in different data sources or even different project reports from the same data source. For example, definition of a class of literate population may have included those who are able to read and write in any language, those who have completed at least basic schooling, or those who have obtained a specific educational certification.

The use of secondary data is significantly influenced by the **recency** of the secondary data. It is difficult to base marketing decisions on data that has been out of date and is no longer useful. We have a decennial census, but the findings are only made public a year or two after the census, significantly restricting the usage of census data, which is a priceless source of extensive information. For example, despite the fact that the 2011 census data provides some of the most important marketing information on income, consumption, expenditure savings, and investment, it would no longer be relevant for the majority of marketing decisions beyond 2015.

5.6 SOURCES OF PRIMARY DATA

Although almost all marketing research problems involve the use of some secondary data, as discussed earlier, both exploratory and conclusive research scenarios demand the use of a significant amount of primary data. Primary data are primarily collected through **respondents**, **analogous case studies**, and **research experiments**.

5.6.1 Respondents

The most significant source of primary marketing data is, by far, respondents. Marketing decisions are distinguished by the fact that they almost always involve, in some way or another, the prediction of the behaviour of market participants, whether they be customers, industrial users, marketing intermediaries, or competitors. Forecasting the behaviour of one or more of the aforementioned categories is necessary for decisions including the introduction of a product, changes to price or distribution channels, allocation of the advertising budget, and reallocation of sales territory. Thus, the majority of marketing research scenarios are characterized by the study of respondents.

The types of data that may be collected from respondents include information on past behaviour, intentions of likely behaviour, level of knowledge, attitudes, and opinions, as well as socioeconomic characteristics and lifestyle data.

Past Behaviour

On the assumption that there is a reasonably consistent relationship between past and future behaviour patterns, past behaviour is widely utilized as a predictor of future behaviour. To gain a sense of how the product or service is consumed, information on respondents' past behaviours is sought, including their exposure to and understanding of communication tools, their consumption habits, and experiences with competitor companies' goods and services, etc.

Intentions

Self-predictions of intended acts to be taken in the future are known as intentions of likely action. They are examples of the respondent's information that is very commonly requested. However, since intention only depicts concurrently valid claims, actual behaviour can be altered by events beyond the respondent's control, such as price adjustments or the arrival of a new, compelling competing brand. The utility of intentions data is much better as a predictive tool, when intentions, given the conditions of purchase can be assigned probabilities.

The decision of a buyer to buy a product or service is affected by their familiarity with it. Another type of data that is requested from respondents is their level of knowledge or awareness. As a marketer of smart television sets, you would like to know what buyers know about your product's attributes, such as picture quality and sound fidelity, additional features offered, price, etc., in comparison to competitors' products, if you believe that these are the factors that have a significant impact on purchase decisions

Attitudes and Opinion

A predisposition mental condition to act in a particular way is represented by attitudes and beliefs. For instance, a respondent's declaration that he prefers freshly ground coffee to instant mixes provides insight into his future actions when faced with the decision to purchase coffee. The most effective applications of attitude research in marketing are in product design, development, and advertising. Information on attitudes and opinions is gathered using both qualitative and quantitative methodologies.

Socio-economic characteristics

Information on some socioeconomic characteristics, such as income, education, occupational status, etc., is sought if there is reason to believe that they are related to the purchase of a good or service in a consumer research situation. Making more exact target market definitions and pricing and advertising decisions may benefit from the study of these kinds of characteristics.

Lifestyles

Using lifestyles to segment markets and reposition products is becoming more common, and targeting, positioning strategies are well matched to the customer profile. The development of lifestyle profiles involves a systematic examination of values, attitudes, opinions, and interests as well as income, all of which can be obtained from respondents using communication methods alone or in conjunction with observation techniques. You already have a basic understanding of the data required and the methods used to construct lifestyle profiles; this was already covered in your MMPC-006 Marketing Management course and also discussed in MMPM-001 Consumer Behaviour.

5.6.2 Primary data from Analogous situations

Case Study

The case study or case history, which emerged from the behavioural sciences, is widely used in modern marketing research. An extensive investigation is conducted to completely examine the case situation using scenarios similar to or pertinent to the problem situation. The focus of the study is on identifying important variables, describing how they relate to one another, and perhaps defining the opportunity or problem in order to recommend different approaches to the decision-making process.

The case study approach is often applicable in circumstances where multiple variables interact to produce the problem or the opportunity. Examples include research problems involving: a) the analysis of how sales performance changes when a new competitor enters the market; b) comparing performance levels in various markets; c) transitional phases, such as the analysis of sales territories where the company is switching its distribution channel from indirect to direct sales.

Simulation

A marketing simulation is a partial depiction of the marketing system that focuses on a certain component. It is mostly a computer-based technique. Through the manipulation of independent variables like the marketing mix or situational factors and the observation of their effects on the dependent variables, the dynamics of the marketing system are studied. As a result, it would require the representation of the phenomenon's properties and the interactions between variables as data inputs.

The researcher building the marketing simulation must conceptualize, record, and assign probability to represent the behaviour of the structural components. These elements might be consumers, retailers, households, etc., depending on the phenomenon being examined, and the factors influencing how these elements would behave could be price charges, advertising expenditure levels, sales promotion programmes, the quality of the product, etc.

The success of simulation, a challenging process, depends on how well reality is reflected in the simulation (abstraction). It is only applicable in limited circumstances while conducting normal marketing research because it calls for specialized expertise.

Experimentation

Experimentation, which is mostly used to investigate cause-and-effect relationships among marketing variables, is also a rather rich source of primary data. Marketing experiments are carried out in a manner that is quite similar to that of other scientific investigations. A conscious manipulation or control of one or more independent variables is employed, and the effects on the dependent variables are investigated. The variables that can make it difficult to discern true causality are controlled to the fullest extent possible. We have already discussed various experimentation design in detail in Unit 4.

5.7 ADVANTAGES AND LIMITATIONS OF PRIMARY DATA

Typically, primary sources offer more in-depth information than secondary sources. This is partially due to the ability to more precisely adjust the data collection methods and tools to the researcher's informational requirements. Additionally, this increases the analysis's adaptability to the current research goal.

The researcher can choose the proper unit for this purpose with a considerably greater degree of fit since he has more flexibility in doing so because terms and units can be specified more accurately. In the case of secondary data, the researcher may not have much control over the choice of units used in the secondary data source because the data have been gathered for some other purpose. When using primary data, the user may assess the level of confidence he may place in the data because he is fully aware of the methodology and tools that were utilised, as well as their strengths and weaknesses.

There are additional benefits that distinguish the various approaches to collecting primary data in addition to the broad benefits mentioned above. The main drawbacks of using primary data include the time and financial expenditures, the lack of qualified interviewers or investigators, and the potential for errors at many stages, such as sample selection, sample size determination, tool selection, etc.

5.8 METHODS OF PRIMARY DATA COLLECTION

When you seek information on something from people in your everyday life, it can only be obtained by either asking them about it or by observing the phenomenon on which it is based. These two different ways of data collection, namely communication and observation, have merely been formalized and systematized by formal research. Within each category, you will find several tools that can be used to affect communication and/or observation. The fact that distinguishes communication methods is that there is a dialogue with the respondent, either spoken or written, while observation methods involve keeping track of or monitoring the behaviour of the subject matter of the research without interfering with it.

5.8.1 Communication methods

Interviews: Interviews in marketing research are by far the most common method of data collection. Interviews may be:

a) **Structured and Direct**- Involving the use of a structured formal questionnaire as well as an interviewer (e.g., surveys using questionnaires).

b) **Unstructured and Direct**- Not involving the use of a pre decided questionnaire, only the interviewer (e.g., personal interviews)

c) **Structured and indirect**- Where the responded to a non-personal ambiguous Data Collection situation which is later interpreted (e.g., projective techniques like word association)

d) **Unstructured and indirect**- where the respondent is asked to respond to non- personal ambiguous situation but where the interviewer has considerable degree of freedom in modifying/altering the situation.

a) Structured and Direct (Surveys using Questionnaires)

Surveys that involve structured questionnaires are the most widely used kind of communication nowadays. It makes sense to ask the people directly if you want to learn about their consumption patterns for a specific product category, the factors influencing their brand choice, or the relative impact of various media on their choices. Formal undisguised questions are predesigned in the form of a questionnaire to obtain comparable information from responders. The questionnaire may be administered with the help of an interviewer, who is instructed to ask the questions exclusively—and only in the order—that are listed on the form.

The two main benefits of employing questionnaires are their adaptability and affordability. The questionnaire approach is adaptable enough to be customized to the requirements of the majority of marketing research situations. Additionally, questioning is the only way to elicit factors such as knowledge, attitudes, intentions, motivations, and personal habits that do not lend themselves to observation. Furthermore, it is the only way to learn about historical occurrences for which records have not been kept.

In comparison to observation, questioning proves to be quicker and less expensive. When employing surveys, interviewers have more control over information gathering than observers, who must wait for the respondent to carry out the action being observed. Additionally, the interviewer can control how long it takes between interviews, while the observer cannot control how long it takes between successive observations.

However, there are several limitations that apply to the questionnaire approach as well. The respondent's inability to provide information is among the most significant of them. Many respondents are actually unable to provide honest answers to the questions even if they may be prepared to provide information about themselves. For instance, you could find it challenging to explain why you choose one brand of soap over another at a certain moment unless you have thoroughly considered these reasons ahead. As a result, on questions about the motivations of the buyer, inaccurate information is frequently provided, interfering with the reliability of the data collected.

Unwillingness of respondent to provide information

The questionnaire-aided interviewer is typically a waste of the respondent's time because he is obliged to answer questions about his socioeconomic status, personal habits, and values to an unidentified interviewer. Respondents occasionally decline to take the time necessary to answer the questions, and frequently they decline to answer inquiries about their income or about very personal matters. These informational gaps may have an impact on the entire research process. reliability of the collected data.

Influence of the questioning process

Given that the situation in which respondents are asked to relate their routine actions is an artificial one, it is possible that the answers they provide will not be accurate. Respondents frequently give responses they believe will satisfy the interrogator. Sometimes, responders have a tendency to give "doctored" or created answers if the real answer to a question puts them in a bad light or is damaging to their ego. Additionally, some respondents attempt the process extremely casually and use their answers to amuse, shock, or astound the interviewer.

b) Unstructured direct interviews

This approach does not use a questionnaire that is formally constructed. Only broad guidelines on the kind of information sought are provided to the interviewer. The interviewer therefore has a lot of freedom in choosing the questions, as well as the wording and order, that appear most relevant for a particular interview. These interviews, which are frequently utilised in exploratory research investigations, are helpful in gaining a greater understanding of the problem and identifying the areas that need more investigation.

This method is also used to learn more about the respondent's intentions behind their behaviour. It typically takes the form of an in-depth interview when used in this context. The interviewer may continue probing questions like "Could you explain what you meant by this statement?" and "Why do you feel that way?" and "Do you have any other reason?" until they have all the information they require while examining the motives behind a certain purchase. The requirements of the problem, the amount of time available, and the respondents' desire and capacity to verbalize their motives are the limitations in this case.

Unstructured direct interviews have the benefit of being done in a conversational, informal style, which is better suited to acquiring information because they are not constrained by formalistic questions. Additionally, the terminology can be changed to ensure compatibility and comprehension. Access to information not often collected by structured questionnaires is frequently made possible by the direct unstructured interview's flexibility and informality. However, there is still a chance that respondents will provide wrong answers, whether knowingly or unknowingly.

The interviewer's skill to craft and ask effective questions is key to the success of the unstructured interview. The cost of each interview increases when highly qualified and competent interviewers are used. Unstructured interviews tend to last longer than structured ones, and the varying order and format of the questions makes editing and tabulating considerably more difficult.

c) Structured and Unstructured indirect interviews

Some techniques have been developed to elicit information in an indirect manner in order to overcome the nonresponse or erroneous response problems associated with direct questionnaires and unstructured interviews. Respondents are asked to use projective techniques developed in the psychology area to depict a non-personal ambigous situation. The assumption is that the respondents' descriptions would represent a projection of their own personalities, values, motivations, wants, and desires. The most common among these techniques are word association tests, sentence completion tests, and thermatic appreception (interpretation of pictorial representation). These techniques have been mostly used in the study of buying motivations and consumption patterns of consumer products that are very similar in performance, price, and quality, so that the consumer has little basis for differentiating them except brand and company image. Examples are soaps and detergents, food products, etc.

All indirect interviews involve some structuring because a predesigned set of words, phrases, or pictorial representations are used to elicit respondents' descriptions. However, the degree might range from highly structured to somewhat unstructured interviews, where the interviewer has a lot of freedom to probe the respondent for more information. You can better understand these tools of data collection by reading the descriptions of some of the most widely used indirect interviewing techniques that follow.

Word association tests

The word association test involves the presentation of a series of stimulus words to a respondent, who is asked to quickly supply the word that first comes to mind after hearing the stimulus word. The respondent would most likely give the word that he most closely associates with the stimulus used. For example, if you are given the name of a specific brand of instant coffee, you may respond in terms of your feelings about its flavor, its evaluation in comparison to ground coffee, its costliness, and its convenience, depending on the dominant association you have for this product. It helps in identifying respondent perceptions about products and services, new packaging designs, advertising campaigns, and so on.

Sentence completion tests

The sentence completion test, which was created as an extension of the word association test, asks the respondent to swiftly finish the sentences after being given their initial phrases. The sentence completion test was employed in a marketing research study to try and determine why people buy cars. The words offered were:

People who drive a Maruti ………………………………………………….

Professionals usually drive …………………………………………………..

Most of the new cars ………………………………………………..

When I drive fast …………………………………………………

The responses given by men and women were found to differ. Men were thought to value fuel economy, mobility, and stylish looks more than women did, who valued safety and comfort more.

The above sentence completion example can be extended by adding similar words for buying car such as

People who drive an Electric car ……………………………………….

Thematic appreciation test (Picture interpretation tests)

The test involves showing one or more pictures or cartoons to the respondent and asking them to describe the situation or assume "the role of" one of the people shown in the picture or cartoon. In most cases, the picture depicts one or more people in an ambiguous situation, and it is anticipated that the respondent will interpret the situation in line with his own personality.

Cartoons involving people and the product under study are frequently used in marketing research situations. Skilled interpreters then use the descriptions offered by the respondents to decipher their associations with the product's price and quality, its mental valuation in comparison to competitors' products, and so on.

The story completion test is another form of this projective technique in which the respondents are given the first few sentences of a situational narrative and asked to complete it.

Focus groups

Focus groups are one of the most widely used type of indirect interview. Here, a group of people jointly participate in an unstructured, indirect interview. In a group discussion exercise, a deliberately chosen group of 8–12 individuals who share a common background and have had similar user experiences with the subject under study are brought together. The interviewer, who plays the role of a moderator, attempts to focus the discussion on the problem areas in a nondirected, relaxed way. The purpose of this exercise is to provide interaction and involvement among the group members doing the interview so that a spontaneous discussion with expressions of attitudes, opinions, values, and the intention of further product use takes place.

Focus groups are more typical of preliminary and exploratory studies than of casual research, and they are designed to collect information to aid in the formulation of hypotheses or the establishment of premises for future investigations. It is useful in research studies that examine potential new product ideas, decide on innovative advertising strategies, generate data for potential product modifications, and identify more effective product or service distribution methods.

Media used in direct and indirect interviews

The interviews can be done in-person, with an interviewer present, or online. Online surveys have the advantage of being less expensive and offering a quick response, and they may be used for both structured and unstructured interviews. Online surveys, however, have a high nonresponse rate.

5.8.2 Observation method

Another type of data collection methods is observation, which involves making and recording observations without engaging with the subjects of the study or the respondents. Sometimes it is less expensive and more accurate to carefully monitor the respondents' behaviour rather than asking them questions. Even though prior behavior cannot be seen, the results of those behaviors can. Instead of asking, "What brand of television do you own?" Is it white or black? Is it controlled remotely? It would be easier to examine the set and watch how it is used.

Observation can be used both alone and in conjunction with other methods to collect data. It is quite often used in exploratory and monitoring research designs. For example, if the newly opened supermarket in your city wanted to know if its prices of commodities were competitive, the only way to get the information was to regularly monitor, through observation, the prices of these commodities in other supermarkets and comparable stores.

Although there are other ways to get data, there are some situations when observation is really a better option due to cost or accuracy issues. In the section on survey limitations, we discussed about respondents' inability or unwillingness to provide accurate information. In this case, observation might prove to be a more useful method of data collection. Studies on brand-buying behaviour have revealed that while respondents underrate taking advantage of discounts, they overrate owning or purchasing a reputable or prestige brand. In these circumstances, observing purchasing trends might produce better outcomes.

Observation may be carried out manually or mechanically. Devices like cameras are used to observe consumer behaviour in a natural setting. Another tool for documenting daily behaviour is the audiometer, which is referred to as the electromechanical version of the respondent's diary. It is installed in a random sample of people's televisions and automatically records the times the set is turned on as well as the stations tuned in. The recordings that are collected are utilised to calculate the audience share and overall average for the various channels and commercials.

Manually, the observation approach includes the use of a trained observer who makes and records firsthand observations of respondent behaviour. Direct observations may be used to get information on the following problems:

a) Who is the actual buyer of the product in question?

b) To what extent do they seem to have a predetermined idea of buying this product?

c) Did they exhibit any signs of being affected by the individual who was with them?

d) How many of them looked up the price or compared it with that of competitors?

e) How many of them read the packaging before making a purchase?

Direct observation just shows the products that were bought, not the justification for those purchases, therefore it only reveals what was bought, not why. Since motives cannot be seen, they must be assumed, and these assumptions are rarely substantiated by reality. The bias of the observer, who might selectively record behaviours that “seem meaningful” to them and ignore others, limits manual observation further. This has effects on the objectivity of the research.

5.9 SOURCES OF ERROR IN PRIMARY DATA COLLECTION

Earlier in this unit, we discussed about sources or error associated with secondary data. Three significant types of errors can also occur during primary data collection procedures. These are **sampling error, non-response error** and **response error**.

Sampling error

Sampling error, as the name suggests, is an error in the selection process that causes the sample to no longer be representative of the population.

For example, selecting a sample of college students from a major city would not be representative of the male population in India if you were researching the smoking habits of Indian men. Regardless of the data collection tool you choose, the study you do on this sample will have sampling error, making it invalid. On the other hand, by changing the properties of the sample taken, the range of sampling error can be modified. Furthermore, if we take a probability sample, we may measure the magnitude of the sampling error.

Non-response error

When a unit included in the sample cannot be reached or has not been reached, it is known as a non-response error. A unit in this context could be an individual, a family, or an establishment. Non-response is likely to happen, for instance, if a proportion of the housewives in a sample from a given city region are absent each time the interviewer decides to visit. As has already been discussed, mail interviews have a very high frequency of non-response error because respondents frequently choose to ignore the questionnaire they receive.

Nonresponse bias is a significant error that occurs in the majority of direct, structured interviews, or surveys that use questionnaires. As families who are not able to be reached after a specific number of efforts during the day may be significantly different from those who can be easily contacted, it may affect both the quality and objectivity of data collection. The non-response error is a big problem because it frequently has no clear direction. Although it is challenging to measure the size of the error, it is possible to establish the maximum error due to non-response by assuming that each non-respondent would have responded in a particular manner.

Response errors

Response errors occur when the reported value of a variable differs from its actual value. We include both communication and observation errors here. We've already discussed two causes of response error: the respondent's inability or unwillingness to provide accurate information due to the time factor, the prestige factor, or the invasion of the primary factor. Let us now discuss the sources of response error related to the **investigator** and the **tools** used by him.

Inaccurate information due to the investigator

The interviewer's dishonesty is the most frequent reason for this kind of error. There are various ways that interviewers willfully collect false information and provide it. If the questionnaire contains a question that the investigator feels awkward to ask, he may choose to provide his own response or make an assumption about what the respondent's response would have been. In rare instances, it was found that interviewers had submitted reports but had never been contacted. Another in-between situation that is discovered to exist is when interviewers ask their own friends and associates to complete the questionnaire or respond to a direct interview while listing the responses in the names of the people listed in the sample, thereby tainting the entire sampling exercise.

Interviewer cheating, according to seasoned marketing research firms, will never be totally eradicated; instead, it can only be regulated to minimize its prevalence. The prevalence of cheating can and does be reduced with careful interviewer selection, training, and supervision. Additionally, various control measures are used to minimize the occurrence of interviewer-generated inaccuracy, such as cross-checking small samples of respondents and using cheater questions, which reveal the fabricated responses with a comparatively high success rate.

Ambiguity

Ambiguity is a source of error that can arise in both communication and observation techniques of data gathering. Ambiguity is described as errors made in interpreting behaviour or words said or written. All languages have the potential to be ambiguous because information is transmitted from one person to another, and the recipient of that information may understand a question or behave in a different way.

5.10 SUMMARY

In this unit, we have discussed the main data types used for marketing research in the context of the various research designs that may be used. The sources of secondary and primary data have been discussed. The two fundamental methods for collecting data, namely communication and observation, have been discussed along with the different variations. The unit ends with a discussion of the sources of error in data collection.

5.11 KEY WORDS

Data Mining: Large datasets are analysed using powerful computers, advanced statistical software, and other tools in data mining.

Data Warehouse: Data from several operating systems in the organizations is consolidated into a single, central database known as a data warehouse.

Non-response Error: When a unit included in the sample cannot be reached or has not been reached, it is known as a non-response error.

Primary Data: refers to the first-hand data gathered by the researcher himself. For example, data collected with the help of questionnaire

Response Errors: Response errors occur when the reported value of a variable differs from its actual value.

Sampling Error: Sampling error, as the name suggests, is an error in the selection process that causes the sample to no longer be representative of the population.

Secondary Data: means data collected and processed by some other agency earlier. For example, Government publications.

5.12 SELF-ASSESSMENT QUESTIONS

1) One of the main benefits of using surveys is that they can provide reliable information about how individuals will behave in the future. Do you agree? justify your answer.

2) Discuss the main sources of primary and secondary data.

3) What is the type of data available from official publications?

4) What are the limitations associated with the use of secondary data?

5) What are the tools of collecting data from respondents?

6) Discuss the important sources of error in both secondary and primary data.

5.13 PROJECT QUESTIONS

Suppose you are planning to open a new retail store that will sell mobile phones. What secondary data is available in your area to help you decide where to locate the store? Would you also plan for primary data collection? Explain.

5.14 FURTHER READINGS

Gopal, M.H. 1964. An Introduction to Research Procedure in Social Sciences, Asia Publishing House: Bombay.

Kothari, C.R. 1989. Research Methodology Methods and Techniques, Wiley Eastern Limited: New Delhi.

Sadhu, A.N. and A. Singh. 1980. Research Methodology in Social Sciences, Sterling Publishers Private Limited: New Delhi.

Wilkinson, T.S. and P.L. Bliandarkar. 1979. Methodology and Techniques of Social Research, Himalaya Publishing House: Bombay.

UNIT 6 DATA PROCESSING

Learning Outcomes

After going through this unit, you should be able to:

- appreciate the importance of data editing
- design the coding scheme for a given data
- classify the data according to attributes or numerical characteristics
- present the data in the forms of tables and graphs

Structure

6.1 INTRODUCTION

Data analysis is the next step after data collection in order to provide answers to our research questions. However, there are several prerequisite tasks that must be finished before we can begin the data analysis. The level of care taken during the data preparation step determines the quality of the outcomes. Results can be seriously impacted by incorrect data preparation, which can result in biased conclusions and inaccurate interpretation. Therefore, ensuring that the data are accurate, complete, and suitable for further research is a crucial stage in the data processing process. Editing, coding, classifying, and tabulating data to make it suitable for data analysis constitute the main components of data processing. These steps of data processing are discussed in details in this unit.

6.2 DATA EDITING

The editing of data is a process of examining the raw data to detect errors and omissions and to correct them, if possible, so as to ensure legibility, completeness, consistency, and accuracy.

The recorded data must be **legible** so that it can be coded later. It needs to ensure, open-ended survey questions in particular are readable and clear to understand. Incorrect responses can be fixed by contacting the individuals who recorded them, or else they can be inferred from other aspects of the question.

The questionnaire must have all of its items properly **completed** in order to be considered complete. When a question is not answered, the interviewers may be contacted to determine whether the candidate did not answer the question or declined to do so. In the first scenario, it's possible that the interviewer won't recall the response. In this situation, the respondent might be contacted once again or, alternatively, this specific piece of information might be considered to be missing.

It is crucial to verify whether the respondent is **consistent** in their responses to the questions. For example, a person who claims to use a credit card may not actually have one.

The survey data may not be **accurate** because of bias or dishonesty on the part of the interviewers. One method to recognise it is to search for a recurring pattern of responses that reflect the interviewer's interviewing style.

This will not only guarantee high-quality data but also make data coding and tabulation easier. In actuality, the editing process entails a close examination of the completed surveys.

The researcher must conduct the editing process at two levels, the first of which is **field editing** and the second of which is **central editing**, to verify that data screening and cleaning, which are essentially the needs of the editing process, have been completed.

Field Editing

The field investigators review the data that was initially gathered in the field to check for any discrepancies, non-responses, illegible responses, or incomplete questionnaires and ensure that any inaccuracies may be corrected as soon as possible. While doing so, care should be taken so that the investigator does not correct the errors of omission by simply guessing what the respondent would have answered if the question was put to him. This type of editing is necessary due to the fact that each individual's writing is different. It varies from individual to individual and is sometimes difficult for the tabulator to understand. Additionally, if the investigator needed to speak with the respondent who filled out the form, it would be much easy to obtain the required justifications. Some researchers randomly interview the same respondents to double-check that the administration process was accurate in order to confirm the validity of the data they have collected. This kind of editing should be carried out as soon as possible after the interview, as it may be necessary sometimes to recollect the memories.

Central Editing

The second stage of editing, commonly known as "in-house editing," is carried out by the researcher. All forms must be meticulously edited by a single person (the editor) for a small field research, or by a small group of individuals for a larger field study. The editor may correct any obvious mistakes, including entries that are placed incorrectly or that are written in daily terms when they should have been in weeks or months, etc. For example, if a scale included seven points, one would discover that someone had chosen a number higher than 7. Another scenario is when someone is asked, "How many days do you travel outside of the city in a month?" and they respond, "35 days." By evaluating the other data included in the schedule, the editor may occasionally be able to record improper or missing responses. If more clarification is required, the respondent may be contacted. It is necessary to remove from the schedules all of the obvious incorrect responses. The editing process requires the editor to be familiar with the guidelines and codes provided to the interviewers. The editor should be the one to start the new (corrected) entry, which should be made in a distinctive manner. For future reference, the schedule should include the date of editing.

Activity 1

A marketing research company is undertaking a survey to find out food consumption habits. As the chief of the computer division, it is your job to edit and analyse the raw data from the surveys. Make a list of the areas you want to concentrate on when editing the raw data.

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6.3 CODING OF DATA

Coding is the process of assigning some symbols (either alphabetical or numeric) to the answers so that the responses can be recorded into a limited number of classes or categories. This is done primarily to make it easier for the researcher to interpret the responses, classify them, and then record the data from the questionnaire. The classes should be appropriate to the research problem being studied. They must be exhaustive and mutually exclusive so that the answer can be placed in only one cell under a certain category. Additionally, each class must be specified in terms of a single concept. Coding is required for an effective analysis of the data. Typically, coding decisions should be taken during the questionnaire's design phase in order to pre-code the most likely answers to each question. This makes it easier to tabulate the data on a computer for later analysis. It should be highlighted that any coding errors should be completely eradicated or at the very least minimised. It takes longer to code for an open-ended question than a closed-ended question. A closed-ended or structured question's coding scheme is fairly straightforward and designed prior to the field work. For example, consider the following question:

What is your Gender?

Male ( )   Female ( )

A male respondent may be assigned a code of "0," while a female respondent may be assigned a code of "1." These codes may be specified prior to the field work, and if the codes are written on all questions of a questionnaire, it is said to be wholly precoded.

The similar method might be applied to categorising numerical data that has either been specified as having relevant categories or has not been categorised at all. For example,

What is your monthly income?

The respondent would provide his monthly income in this case, which could be placed in the appropriate column. The same query may alternatively be asked in the following manner:

What is your monthly income?

< Rs. 15000

Rs. 15000 - 19999

Rs. 20000 – 24999

Rs. 25000 or above.

We may code the class less than Rs.15000' as, 1', Rs. 15000 - 19999' as `2', `Rs. 20000 - 24999' as `3' and `Rs. 25000 or above' as `4'.

Open-ended questions are more challenging to code because the interviewer captures the respondent's full comment. What categories are these responses to be placed in? The researcher may select at random 80–90 answers to a question and list them. The coding technique for categorised data as stated above is utilised after deciding which categories are appropriate to employ to summarise the data after looking at the list. A word of caution: when categorising the data, we should give space for "any other" in order to accommodate responses that might not fit into our predetermined categories.

The requirement that the response categories be mutually exclusive and collectively exhaustive may be kept in mind.

A study on consumer awareness and purchase intention for organic food products was undertaken by Poonam Porwal and Rajeev K Shukla with the following objectives:

- To assess the level of consumer awareness of organic products
- To identify factors that influences consumers' intention to purchase organic products.
- To know how consumer demographics affect their purchase intention for organic products.

A questionnaire was designed to achieve these objectives. A sample of the questionnaire is provided below, along with a discussion of its coding structure. Please keep in mind that the aim of this exercise is to create a coding scheme for any particular study questionnaire, not to analyse the questionnaire itself.

Questionnaire

1. **Demographic Profile of Respondents**

   Please fill out the following details**:**

   a) **Age** (in years) Below18 ( ), 18- 25 ( ), 26-35 ( ), 36-45 ( ), Above 45 ( )

   b) **Gender** Male ( ) Female ( )

   c) **Marital Status** Unmarried ( ) Married ( )

   d) **Education** Under graduate ( ), Graduate ( ), Post graduate ( ), Others ( )

   e) **Monthly Income** Below 20,000 ( ), 20,001-40,000 ( ), above 40,000 ( )

   f) **Occupation** Student ( ), Service ( ), Business ( ), Housewife ( ), Others ( )

2. **Based on your understanding about organic food products, rate the following statement on five-point likert scale**

   (1) Strongly Disagree to (5) Strongly Agree – [Strongly agree (5) Agree (4) can't say (3) Disagree(2) Strongly disagree (1)]

Statements	5	4	3	2	1
Organic products have more freshness					
Organic products have superior quality					
Organic food are natural food products					
Organic product is tastier					
Organic products are affordable					
Organic products are for the elite class					
Organic products are for everyone					
Organic products are safe					
Organic products are easily available					
Organic products are for the young people					
Organic products are traditional					
Organic products are trendier					
Organic products are costly					
Organic food has more nutritional value than conventional food products					
I would buy organic food products					
I would like to purchase organic food products even it offered at higher price					

Question number 1 is divided into six parts. For each of the part one separate column is required. 1(a) indicates the age of the respondent. Here age has been put in categorized form. If the respondent has an age below 18 years, the response is coded as 1, if age is between 18-25 years, it is coded as 2, in case it is between 26-35; the code is 3. From 36-45; the code is 4, and above 45 years; the code is 5. Question 1(b) is concerning the gender of the respondent. Here male respondents are coded as 1 whereas female respondents are coded as 2. Question 1 (c) indicates the marital status of the respondents. Here unmarried respondents are coded as 1 whereas married respondents are coded as 2. Question 1 (d) is concerned with educational qualification of the respondents. Here undergraduate respondents are coded as 1, graduates are coded as 2, post graduate as 3, and others responses are coded as4. Question 1(e) mention the monthly income of the respondents put in categorized form. Here the responses are mutually exclusive and collectively exhaustive. If the respondent has a monthly income of less than 20000 rupees, the response is coded as 1, if monthly income is between 20001-40000 rupees, it is coded as 2, and above 40000; the code is 3. Question 1f is concerned with the occupation of the respondent. Here responses from students are coded as 1, service persons are coded as 2, responses of business persons are coded as 3, housewife responses as 4, and others as 5. Question 1 (f) is concerned with the occupation of the respondents. Here undergraduate respondents are coded as 1, graduates are coded as 2, and others responses are coded as 3.

Some of the attributes of organic food products are mentioned in question number 2 and the respondent is assigned the job of rating each of them on a scale of 1 to 5.

The data sheet corresponding to the above coding scheme is shown in the table 6.1. Total 162 responses were received for the study. However, sample format of 20 responses on some attributes are shown here in the table 6.1.

Table 6.1 Data Sheet for the Coded Sample Questionnaire

Respo nden	Age	Gen der	Marit alsts	Educ ation	In come	Occu pation		Sq. Qlty	nat	tas ty		eli te	every one		availa bility	yo ung	tradi tional	tre ndy	cos tly
1	4	1	2	4	3	3	1	5	4	5	4	4	1	2	3	5	4	4	
2	4	2	2	2	3	5	4	5	4	5	3	5	3	5	3	5	1	3	
3	1	1	1	1	3	1	5	5	5	5	1	4	3	5	3	4	1	5	
4	4	1	2	3	1	3	5	5	5	5	2	2	5	5	3	5	5	5	
5	1	1	1	1	1	1	5	5	4	4	4	4	4	5	5	4	4	4	
6	1	1	1	1	2	1	5	5	4	4	4	1	5	4	5	5	5	4	
7	5	1	2	2	1	3	5	5	5	4	3	4	5	4	2	2	2	3	
8	2	1	1	1	1	1	5	5	5	5	4	3	4	5	2	3	3	4	
9	5	2	2	3	1	2	4	5	4	3	2	3	1	4	2	3	3	3	
10	2	1	1	1	1	2	5	5	5	4	2	4	5	5	3	4	3	4	
11	5	2	2	3	2	5	3	4	4	3	2	1	4	4	2	2	5	4	
12	2	1	1	2	1	2	4	4	4	4	3	1	5	4	1	4	4	3	
13	5	1	2	2	2	3	4	4	4	4	3	1	5	5	1	4	4	3	
14	2	2	1	2	1	1	4	5	4	5	4	5	4	5	5	2	2	3	
15	5	1	2	2	3	2	3	5	3	3	4	2	4	4	2	2	3	3	
16	3	1	2	3	3	3	5	5	5	3	1	5	1	3	1	1	4	5	
17	2	1	1	3	3	3	3	4	5	3	2	1	5	5	2	1	4	4	
18	2	1	1	1	2	3	4	5	5	5	5	2	5	5	1	2	4	4	
19	3	1	2	4	3	3	5	5	5	3	3	1	3	3	2	3	3	3	
20	5	2	2	2	1	3	5	5	5	5	5	2	5	5	2	1	5	2	

Activity 2

Describe the demographic profile of 10th respondent as given in data sheet of Table 6.1

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6.4 CLASSIFICATION OF DATA

In most research studies, voluminous raw data collected through a survey needs to be reduced into homogeneous groups for any meaningful analysis. This necessitates the classification of data, which in simple terms is the process of arranging data in groups or classes on the basis of some characteristics. Classification condenses data, allows for comparisons, aids in the study of relationships, and aids in the statistical treatment of data. The classification should be unambiguous, mutually exclusive, and collectively exhaustive. Furthermore, it should be not only flexible but also appropriate for the purpose for which it is sought. Classification can be done using **attributes** or **numerical** characteristics.

1. **Classification According to Attributes:**

   To classify the data according to attributes, we use descriptive characteristics like sex, caste, education, user of a product, etc. The descriptive characters are the ones that cannot be measured quantitatively. One can only talk in terms of its presence or absence. The classification according to attributes may be of two types:

   i) **Simple Classification**: In simple classification, each class is divided into two subclasses, and only one attribute is studied, such as whether the person is a product user or not, married or unmarried, employed or unemployed, and so on.

   ii) **Manifold Classification**: In the case of manifold classification, more than one attribute is considered. For example, the respondents in a survey may be classified as users of a particular brand of a product and non-users of that brand of product. Both users and non-users can be further classified into males and females. Furthermore, males and females can be divided into two groups: those under the age of 25 and those aged 25 and up. We can further classify them as professionals or non-professionals. This way, one can keep on adding more attributes. This is shown in Figure 6.1. However, the addition of a particular attribute (the process of sub-classification) depends upon the basic purpose for which the classification is required.

The objectives of such a classification have to be clearly spelled out.

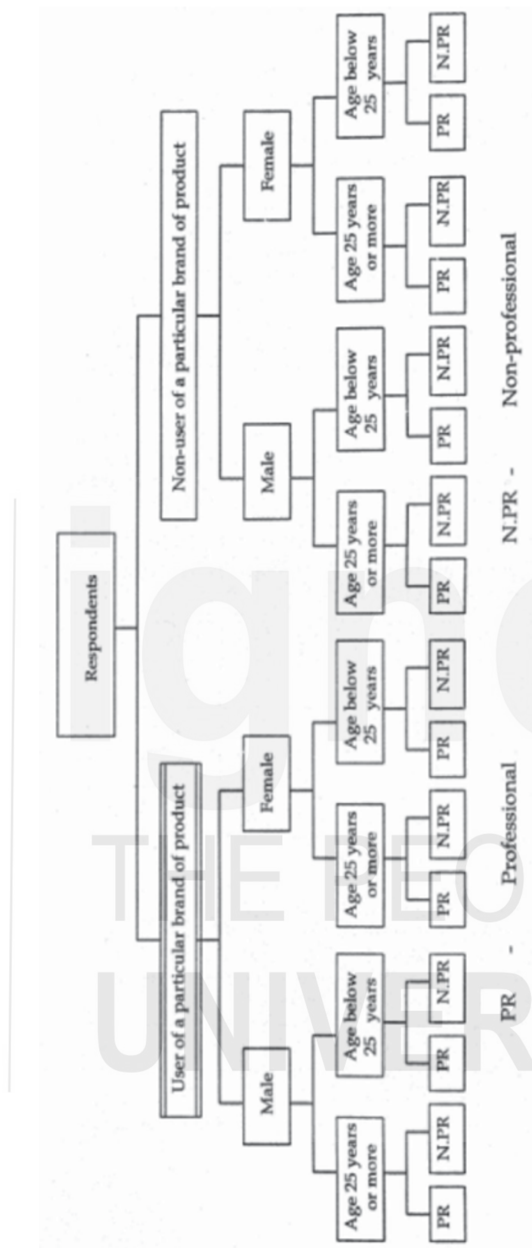

Figure 6.1 An Example of Manifold Classification

2. **Classification According to Numerical Characteristic:**

When the observations possess numerical characteristics such as sales, profits, height, weight, income, and marks, they are classified according to class intervals. For example, persons whose monthly income is between Rs. 12001 and Rs. 15000 may form one group; those whose income is between Rs. 25001 and Rs. 30000 may form another group; and so on. In this manner, the entire set of data may be divided into a number of groups or classes, which are usually called class intervals. The number of items in each class is called the frequency of the class. Every class has two limits: an upper limit and a lower limit, which are known as class limits. The difference between these two limits is called the magnitude of the class or the width of the class interval. The class intervals may be formed by using the inclusive and exclusive methods. Assume we have class intervals like 10—15, 16—21, 22—27, and so on. Such a class interval is an example of the inclusive method because both the lower and upper limits are included in the class. If the variable X falls in the first-class interval, it can take values like $10 = X \leq 15$. The class intervals like 10 - 15, 15 - 20, 20 - 25 etc. form an example of exclusive class interval since the lower limit is included whereas the upper limit is excluded from the class interval. If the variable X falls within the first-class interval, its values would be $10 = X < 15$. As an illustration of how the data can be classified into class intervals using the inclusive and exclusive methods, we may consider the following example.

Example: The following data refers to the sales of a company for the past 40 quarters. Tabulate the data using the inclusive method.

060,	1310,	1255,	750,	1690,	945,	1200,	106
125,	2120,	1190,	1120,	2130,	2240,	2190,	137
440,	2560,	870,	2000,	1870,	1700,	1800,	237
940,	2250,	1460,	1750,	1875,	1165,	2255,	147
060,	2135,	2125,	1760,	1650,	1945,	2000,	225

We will be using the data given above. We form five class intervals, each of width 370. These are inclusive class intervals in the sense that the variable X could take any value between the lower and upper limits in such a way that both ends of the interval could be covered by this. The class intervals along with the number of items in each class interval are shown in the Table 6.2 below:

Table 6.2 Classification by using Inclusive method

Class Intervals	Tally Marks	Frequencies
750-1120	卌 I	6
1121-1491	卌 IIII	9
1492-1862	卌 I	6
1863-2233	卌 卌 III	13
2234-2604	卌 I	6

Activity 3

A survey was conducted to estimate the expenditure of households on entertainment. The data for a number of variables was collected. One of the variables of interest is monthly income. The data for the monthly household income (Rs. thousand) of the sample of 50 respondents is given below. Use the data to form a class interval using the exclusive method.

20	12	15	27	28	40	42	35	37	43
55	65	53	62	29	64	69	36	25	18
56	55	43	35	26	21	48	43	50	67
14	23	34	59	68	22.	41	42	43	52
60	26	26	37	49	53	40	20	18	17

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6.5 TABULATION AND GRAPHICAL PRESENTATION OF DATA

Tables and graphs can be used to present the data that has been collected. Data in tabular form is classified according to time or another variable, and graphs or charts are used as a visual form of data presentation. The tabulation is used for summarization and condensation of data. It aids in the analysis of relationships, trends, and other summarization of the given data.

Following are the important characteristics of a table:

i) Every table should have a clear and concise title to make it understandable without reference to the text. This title should always be just above the body of the table.

ii) Every table should be given a distinct number to facilitate easy reference.

iii) Every table should have captions (column headings) and stubs (row headings) and they should be clear and brief.

iv) The units of measurements used must always be indicated.

v) Source or sources from where the data in the table have been obtained must be indicated at the bottom of the table.

vi) Explanatory footnotes, if any, concerning the table should be given beneath the table along with reference symbol.

vii) The columns in the tables may be numbered to facilitate reference.

viii) Abbreviations should be used to the minimum possible extent.

ix) The tables should be logical, clear, accurate and as simple as possible.

x) The arrangement of the data categories in a table may be a chronological, geographical, alphabetical, or according to magnitude to facilitate comparison.

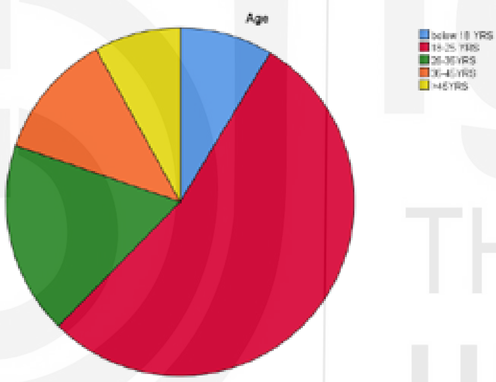
xi) Finally, the table must suit the needs and requirements of the research study.

In order to present statistical data, a variety of graphs and charts are used. Bar charts, two-dimensional diagrams, pictograms, pie charts, and arithmetic or line charts are among those that are frequently used.

These aspects of tabulation and graphical presentation have been widely covered in Unit 2 of the course MMPC-005: Quantitative Analysis for Managerial Applications (https://egyankosh.ac.in/handle/123456789/79288). You may therefore refer to the said study material. However, the tables and charts relating to the data collected using the questionnaire earlier discussed in the coding section and presented on Table 6.1 are illustrated here by taking into consideration the total responses (162) obtained for your quick reference.

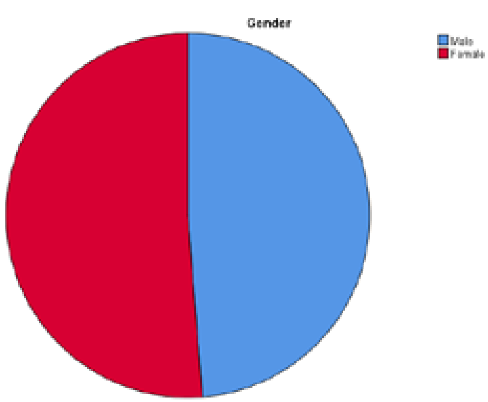
Age

	Frequency	Percent	Valid Percent	Cumulative Percent
below 18 YRS	14	8.6	8.6	8.6
18-25 YRS	87	53.7	53.7	62.3
26-35YRS	29	17.9	17.9	80.2
36-45YRS	19	11.7	11.7	92.0
>45YRS	13	8.0	8.0	100.0
Total	162	100.0	100.0	



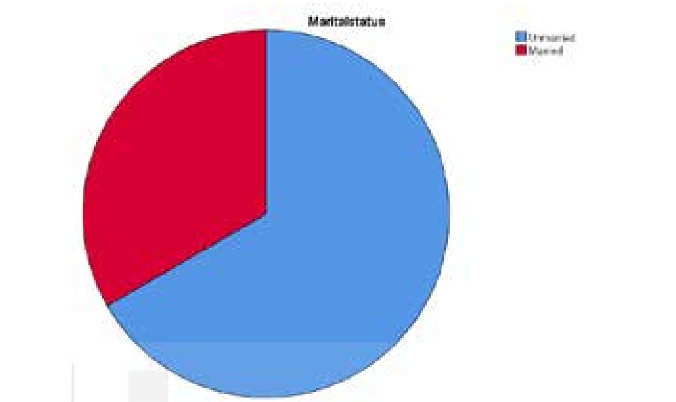
Gender

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Male	79	48.8	48.8	48.8
	Female	83	51.2	51.2	100.0
	Total	162	100.0	100.0	



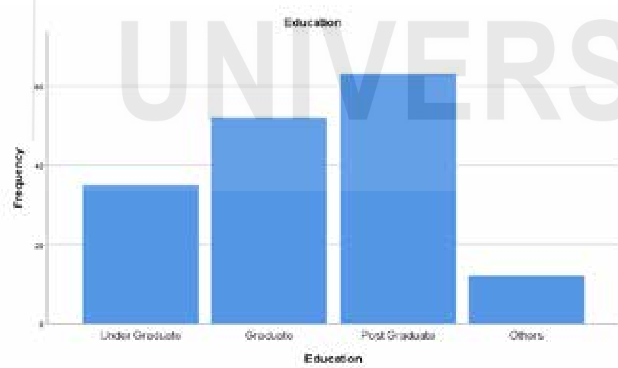
Marital status

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Unmarried	108	66.7	66.7	66.7
	Married	54	33.3	33.3	100.0
	Total	162	100.0	100.0	



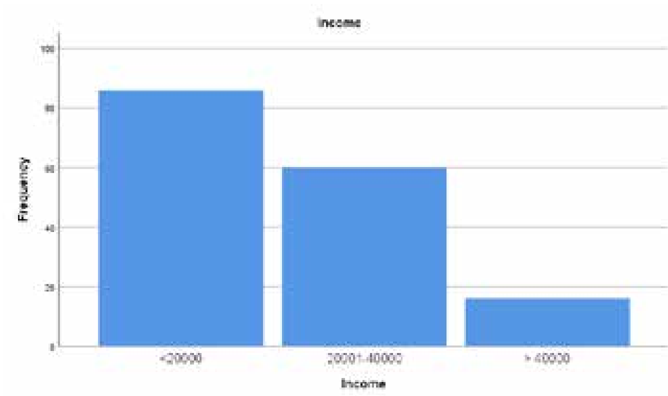
Education

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Under Graduate	35	21.6	21.6	21.6
	Graduate	52	32.1	32.1	53.7
	Post Graduate	63	38.9	38.9	92.6
	Others	12	7.4	7.4	100.0
	Total	162	100.0	100.0	



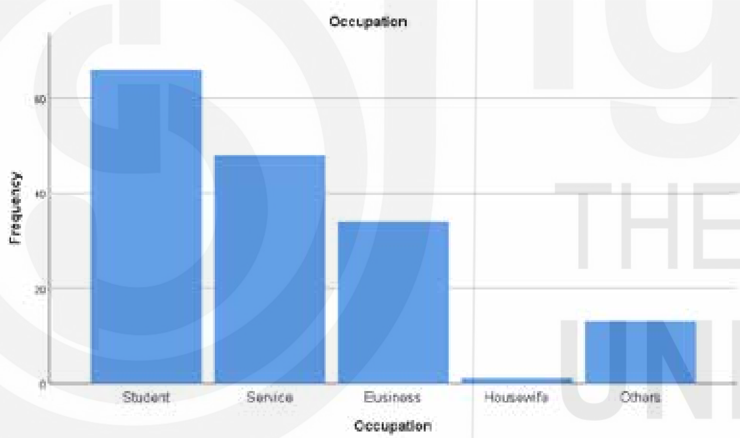
Income

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	<20000	86	53.1	53.1	53.1
	20001-40000	60	37.0	37.0	90.1
	> 40000	16	9.9	9.9	100.0
	Total	162	100.0	100.0	



Occupation

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Student	66	40.7	40.7	40.7
	Service	48	29.6	29.6	70.4
	Business	34	21.0	21.0	91.4
	Housewife	1	.6	.6	92.0
	Others	13	8.0	8.0	100.0
	Total	162	100.0	100.0	



Activity 4

Using your own data source, present the data in tabular form and draw an appropriate chart to present the same.

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6.6 SUMMARY

This unit has discussed various aspects of data processing, including editing, coding, classification, and the presenting of data in tables and graphs. There are two types of editing: central editing and field editing. Coding is the process of assigning specific symbols, numbers, or both to the responses to the questions in a questionnaire so that the responses can be recorded in a finite number of classes or categories. This facilitates data analysis. The design of a coding scheme has been discussed with the help of a sample questionnaire. Classification is the process of arranging data into groups or classes on the basis of certain characteristics. It involves the condensation of data, which facilitates comparison and helps in establishing relationships between variables. Classification can be based on attributes or numerical characteristics. The former can be classified as simple or manifold classification. The latter is achieved using either an inclusive or an exclusive method of forming frequency distribution. The data may be presented in the form of tables or graphs.

6.7 KEY WORDS

Classification of Data: is the process of arranging data in groups or classes on the basis of some characteristics.

Coding of Data: Coding is the process of assigning some symbols (either alphabetical or numeric) to the answers so that the responses can be recorded into a limited number of classes or categories.

Data Editing: The editing of data is a process of examining the raw data to detect errors and omissions and to correct them, if possible, so as to ensure legibility, completeness, consistency, and accuracy.

Graphical Presentation: Exhibition of the data in a graph or diagram.

Tabulation of Data: The tabulation is used for summarization and condensation of data.

6.8 SELF-ASSESSMENT QUESTIONS

1. Describe, in brief, the importance of editing and coding of data in the context of a research study.

2. Discuss with the help of suitable examples various steps involved in data processing.

3. Discuss the different aspects of the classification of data. What are the likely problems encountered during classification, and how can they be resolved?

4. Why is tabulation considered essential in a research study? Describe the characteristics of a good table.

5. The following table shows the monthly pay for 32 sales personnel at a company. After deciding on an appropriate class interval, tabulate the data.

12250, 11800, 11650, 11760, 13520, 15600, 12450, 12680
12700, 11680, 13650, 13240, 15850, 13150, 11860, 12425
14520, 13275, 14215, 13760, 11950, 11850, 13750, 12825
14500, 13800, 14300, 12750, 14370, 13350, 12375, 13215

6.9 PROJECT QUESTIONS

Indicate the diagrams you would consider most appropriate to use for representing each of the following classes of statistical data stating briefly the reason for choice:

a) The distribution of a large number of candidates according to the number of marks scored by each in any entrance examination.

b) Total value of India Exports and Imports during the years 2021-2022.

6.10 FURTHER READINGS

Beri, G. C. “Marketing Research - Text and Cases” Tata McGraw-Hill Publishing Co. Ltd.

Kinnear, Thomas C. and James R. Taylog, “Marketing Research - An Applied Approach” McGraw-Hill International Editions (3rd Edition).

Luck, David J. and Ronald S. Rubin, “Marketing Research “ Prentice-Hall of India Pvt. Ltd.

Malhotra, N.K., Marketing Research: An Applied Orientation, seventh edition, Pearson Educations.

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